Chapter 14
Fast Ethernet Controller (FEC)

14.1 Overview

The fast Ethernet controller (FEC) is an ethernet MAC plus two 1 Kbyte FIFOs that work under the control of the processor and BestComm DMA engine to support 10/100 Mbps Ethernet/802.3 networks. Table 14-1 shows a block diagram.

A brief introduction and overview of the major functional blocks aid in understanding and programming the FEC.

The FEC is controlled by writing through the system interface (SIF) module into control registers located in each block. The control/status register (CSR) block provides global control and interrupt handling registers. User programming of the CSR is the primary focus of this chapter.

The RISC based controller provides the following functions:

- Initialization
- Address recognition for receive frames
- Random number generation for transmit collision backoff timer

The FIFO controller is the focal point of all data flow in the FEC. The FIFO is divided into a transmit and receive FIFO of 1Kbyte each. Transmit data flows from the CommBus into the transmit FIFO and through the transmit block to the physical layer device (PHY). Receive data flows from the PHY to the receive block and is pulled out of the FIFO by BestComm. BestComm data transfers are interrupt driven. Interrupt driven data movement from the processor is not supported.

The bus controller decides which block is to be the T-bus master for each cycle. All the blocks receive their control information over the T-bus and, for the most part, provide status information over this same internal bus.

The media independent interface (MII) block provides a serial channel for control/status communication with the external physical layer device (transceiver or PHY). The serial channel consists of the MDC (clock) and MDIO (bidirectional data I/O) lines of the MII interface.

The transmit and receive blocks provide the ethernet MAC functionality (with some assistance from the microcode). Internal to these blocks are clock domain boundaries between the system clock and the network clocks supplied by the PHY.

The management information base (MIB) block maintains the counters for a variety of network events and statistics. The counters support the RMON (RFC 1757) ethernet statistics group and some of the IEEE 802.3 counters.

The FEC supports several standard MAC-PHY interfaces to connect to an external ethernet transceiver. One is the 10/100 Mbps MII interface. Another is the 10-Mbps only 7-Wire interface, which uses a subset of the MII pins.

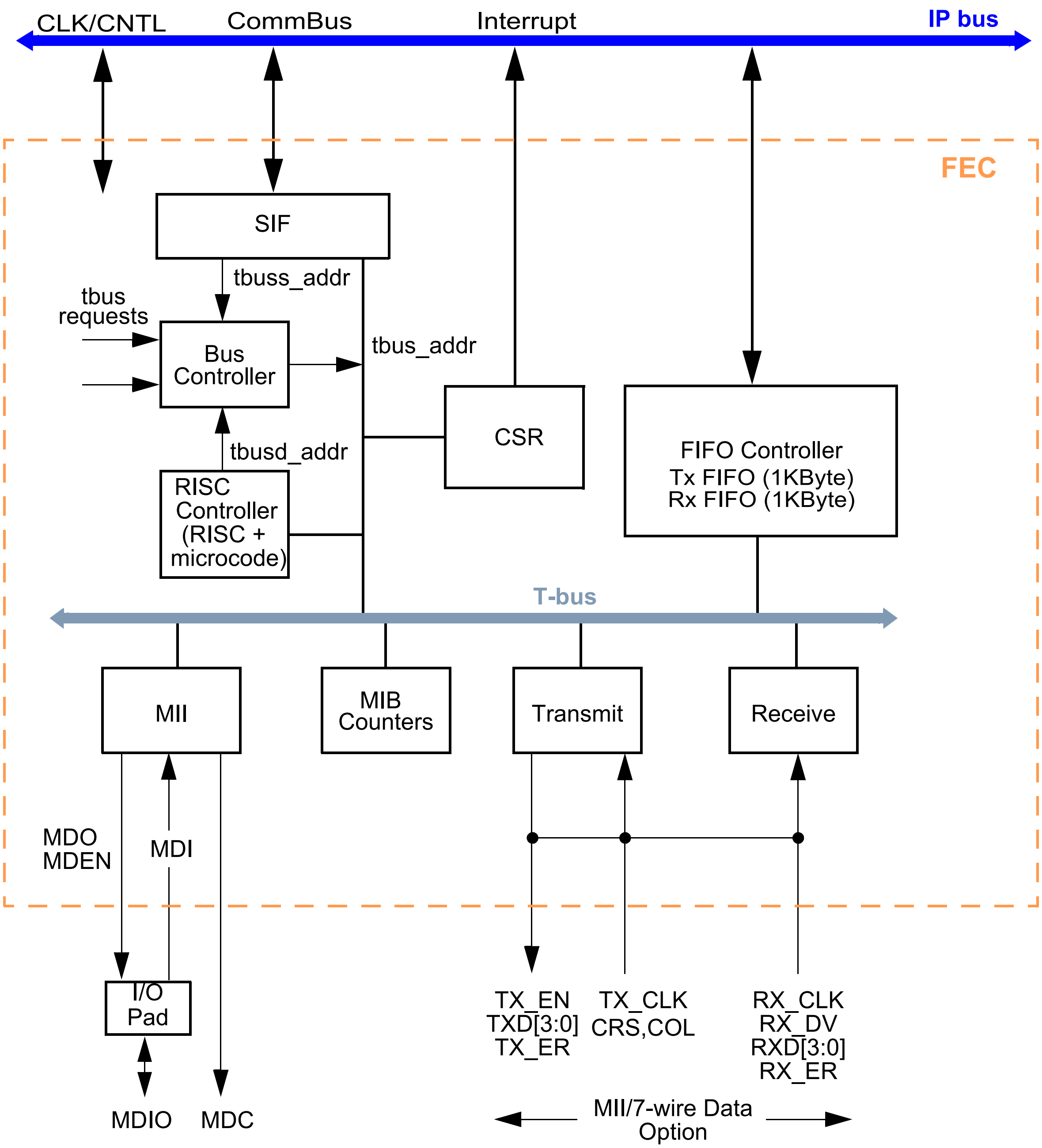


Figure 14-1. Block Diagram—FEC

14.1.1 Features

The FEC incorporates several features/design goals that are key to its use:

- Support for different ethernet physical interfaces:
  - 100 Mbps IEEE 802.3 MII
  - 10 Mbps IEEE 802.3 MII
  - 10 Mbps 7-wire interface (industry standard)
- IEEE 802.3 full-duplex flow control
- Programmable max frame length supports IEEE 802.1 VLAN tags and priority
- Support for full-duplex operation (200 Mbps throughput) with a minimum system clock rate of 50 MHz.
- Support for half-duplex operation (100 Mbps throughput) with a minimum system clock rate of 25 MHz.
- Large (1 Kbyte) on-chip transmit and receive FIFOs to support a variety of bus latencies.
- Retransmission from transmit FIFO following a collision (no processor bus utilization).

- Automatic internal flushing of the Rx FIFO for runts (collision fragments) and address recognition rejects (no processor bus utilization).
- Address recognition
  - Frames with broadcast address may be always accepted or always rejected
  - Exact match for single 48-bit individual (unicast) address
  - Hash (64-bit hash) check of individual (unicast) addresses
  - Hash (64-bit hash) check of group (multicast) addresses
  - Promiscuous mode

14.2 Modes of Operation

The primary operational modes are described in this section.

14.2.1 Full- and Half-Duplex Operation

This is determined by the X_CNTRL register FDEN bit. Full-duplex mode is intended for use on point to point links between switches or end node to switch. Half-duplex mode is used in connections between an end node and a repeater or between repeaters.

Full-duplex flow control is an option that may be enabled in full-duplex mode.

14.2.2 10 Mbps and 100 Mbps MII Interface Operation

The MAC-PHY interface operates in MII mode by asserting the R_CNTRL register MII_MODE bit. MII is the media independent interface defined by the 802.3 standard for 10/100 Mbps operation.

Speed of operation is determined by the TX_CLK and RX_CLK pins, which are driven by the transceiver. The transceiver either auto-negotiates the speed or it may be controlled by software using the serial management interface (MDC/MDIO pins) to the transceiver.

14.2.3 10 Mbps 7-Wire Interface Operation

If the external transceiver supports 10 Mbps only and uses a 7-wire style interface then deassert the R_CNTRL register MII_MODE bit in the R_CNTRL register. This style of interface is not defined by the 802.3 standard, but instead is an industry standard.

14.2.4 Address Recognition Options

The options supported are promiscuous, broadcast reject, individual address hash or exact match and multicast hash match. Refer to the R_CNTRL register for address recognition programming.

14.2.5 Internal Loopback

Internal loopback mode is selected using the R_CNTRL register LOOP bit.

14.3 I/O Signal Overview

This section defines the FEC-to-chip pin I/O. The FEC network interface supports multiple options. One is the MII option that requires 18 I/O pins and supports both data and an out-of-band serial management interface to the PHY (transceiver) device. The MII option supports both 10 and 100 Mbps ethernet rates. The second is referred to as the 7-wire interface and supports only 10 Mbps ethernet data. The 7-wire interface uses a subset of the MII signals.

Table 14-1 shows the network interface signals and lists 18 signals, all of which are used for the 10/100 MII interface.

NOTE

The MDIO pin is bidirectional and corresponds to the FEC block MDI, MDO and MDIO pins. The 7-wire interface option uses a subset of these signals.

Table 14-1. Signal Properties

Signal Name	Chip Pin	Function	Reset State
tx_en	ETH0	MII—transmit data valid output 7-wire—transmit data valid output	0
tdata[0]	ETH1	MII—transmit data bit 0 output 7-wire—transmit data output	

Table 14-1. Signal Properties (continued)

Signal Name	Chip Pin	Function	Reset State
tdata[1]	ETH2	MII—transmit data bit 1 output	
tdata[2]	ETH3	MII—transmit data bit 2 output	
tdata[3]	ETH4	MII—transmit data bit 3 output	
tx_er	ETH5	MII—transmit error output	0
mdc	ETH6	MII—management clock output	0
mdi mdo md_en	ETH7	MII—management data bidirect	Hi-Z (input)
rx_dv	ETH8	MII—Rx data valid input 7-wire—rena input	
rx_clk	ETH9	MII—Rx clock input 7-wire—Rx clock input	
col	ETH10	MII—collision input 10 Mbps 7-wire—collision input	
tx_clk	ETH11	MII—transmit clock input 7-wire—transmit clock input	
rdata[0]	ETH12	MII—Rx data bit 0 input 7-wire—Rx data input	
rdata[1]	ETH13	MII—Rx data bit 1 input	
rdata[2]	ETH14	MII—Rx data bit 2 input	
rdata[3]	ETH15	MII—Rx data bit 3 input	
rx_er	ETH16	MII—Rx error input	
crs	ETH17	MII—carrier sense input	

14.3.1 Detailed Signal Descriptions

14.3.1.1 MII Ethernet MAC-PHY Interface

This section gives a detailed description of the Media-Independent Interface (MII). An overview of the MII is presented followed by a description of the MII signals. Two different types of MII frames are described. A brief MII management function overview is given.

The MII interface has 18 signals. Tx and Rx functions require 7 signals each:

- 4 data signals
- 1 delimiter
- 1 error
- 1 clock

Media status is indicated by 2 signals:

- 1 signal indicates a carrier is present.
- 1 signal indicates a collision occurred.

Management interface is provided by 2 signals.

MII signals are described below.

Tx_CLK . . . . A continuous clock that provides a timing reference for Tx_EN, TxD, and Tx_ER. The frequency of Tx_CLK is 25% of the transmit data rate, ± 100 ppm. Duty cycle shall be 35%-65% inclusive.

Rx_CLK . . . . A continuous clock that provides a timing reference for Rx_DV, RxD, and Rx_ER. The frequency of Rx_CLK is 25% of the Rx data rate, with a duty cycle between 35% and 65%.

Tx_EN . . . . Assertion of this signals indicates valid nibbles are being presented on the MII. This signal is asserted with the first nibble of preamble and is negated prior to the first Tx_CLK following the final nibble of the frame.

TxD. . . . . TxD[0:3] represent a nibble of data when Tx_EN is asserted and have no meaning when Tx_EN is de-asserted. Table 14-2 summarizes the permissible encoding of TxD.

Tx_ER . . . . Assertion of this signal for one or more clock cycles while Tx_EN is asserted causes PHY to transmit one or more illegal symbols. Asserting Tx_ER has no affect when operating at 10 Mbps or when Tx_EN is de-asserted This signal transitions synchronously with respect to Tx_CLK.

Rx_DV . . . . When this signal is asserted, PHY is indicating a valid nibble is present on the MII. This signal remains asserted from the first recovered nibble of the frame through the last nibble. Assertion of Rx_DV must start no later than the SFD, and exclude any EOF.

RxD. . . . . RxD[0:3] represents a nibble of data to be transferred from the PHY to the MAC when Rx_DV is asserted. A completely formed SFD must be passed across the MII. When Rx_DV is not asserted, RxD has no meaning. There is an exception to this which is explained later. Table 14-3 summarizes the permissible encoding of RXD.

Rx_ER . . . . When Rx_ER and Rx_DV are asserted, the PHY has detected an error in the current frame. When Rx_DV is not asserted, Rx_ER shall have no affect. This signal transitions synchronously with Rx_CLK

CRS . . . . Signal is asserted when Tx or Rx medium is not idle. If a collision occurs, CRS remains asserted through the duration of the collision. This signal is not required to transition synchronously with Tx_CLK or Rx_CLK.

COL . . . . Signal is asserted on a collision detection, and remains asserted while the collision persists. The signal behavior is not specified when in full-duplex mode. This signal is not required to transition synchronously with Tx_CLK or Rx_CLK.

MDC. . . . . Signal provides a timing reference to the PHY for data transfers on the MDIO signal. MDC is aperiodic, and has no maximum high or low times. The minimum high and low times is 160 ns, with the minimum period being 400 ns.

MDIO. . . . . Signal transfers control/status information between the PHY and MAC. It transitions synchronously to MDC. The MDIO pin is a bidirectional pin. The internal FEC signals that connect to this pad are: MDI (data in), MDO (data out), and MD_EN (direction control, high for output).

Table 14-2 lists the interpretation of possible encodings for Tx_EN and Tx_ER

Table 14-2. MII: Valid Encoding of TxD, Tx_EN and Tx_ER

TX_EN	TX_ER	TXD	Indication
0	0	0000 through 1111	Normal inter-frame
0	1	0000 through 1111	Reserved
1	0	0000 through 1111	Normal data transmission
1	1	0000 through 1111	Transmit error propagation

A false carrier condition occurs if PHY detects a bad start-of-stream delimiter. This condition signals MII by asserting Rx_ER and placing 1110 on RxD. Rx_DV must also be de-asserted. Valid Rx_DV, Rx_ER and RxD[3:0] encodings are shown in Table 14-3.

Table 14-3. MII: Valid Encoding of RxD, Rx_ER and Rx_DV

RX_DV	RX_ER	RXD	Indication
0	0	0000 through 1111	Normal inter-frame
0	1	0000	Normal inter-frame
0	1	0001 through 1101	Reserved
0	1	1110	False Carrier
0	1	1111	Reserved
1	0	0000 through 1111	Normal data reception
1	1	0000 through 1111	Data reception with errors

14.3.1.2 MII Management Frame Structure

A transceiver management frame transmitted on the MII management interface uses the MDIO and MDC pins. A transaction or frame on this serial interface has the following format:

<preamble><st><op><phyad><regad><ta><data><idle>

Table 14-4. MMI Format Definitions

Name	Description
<preamble>	Optional—consists of a sequence of 32 continuous logic 1s.
<st>	Start of frame—indicated by a <01> pattern.
<op>	Operation code: Read instruction is <10> Write instruction is <01>
<phyad>	A 5-bit field that lists up to 32 PHYs be addressed. The first address bit transmitted is the msb of the address.
<regad>	A 5-bit field that lets 32 registers be addressed within each PHY. The first register bit transmitted is the msb of the address.
<ta>	A 2-bit field that provides spacing between the register address field and the data field to avoid contention on the MDIO signal during a read operation.
<data>	Data field is 16 bits wide. Data bit 15 is first bit transmitted and received.
<idle>	During idle condition, MDIO is in the high impedance state.

14.3.1.2.1 MII Management Register Set

The MII management register set located in the PHY may consist of a basic register set and an extended register set as defined in Table 14-5.

Table 14-5. MII Management Register Set

Register Address	Register Name	Basic/Extended
0	Control	B
1	Status	B
2:3	PHY Identifier	E
4	Auto-Negotiation Advertisement	E
5	AN Link Partner Ability	E
6	AN Expansion	E
7	AN Next Page Transmit	E
8:15	Reserved	E
16:31	Vendor Specific	E

14.4 FEC Memory Map and Registers

The FEC device is programmed by a combination of control/status registers (CSRs) and BestComm task loops. Since the FEC software model is BestComm-based, there is no similarity with existing CPM-based products' coding.

The CSRs are used for mode control, interrupts and extraction of status information. BestComm tasks are used to pass data buffers and related buffer or frame information between the hardware and software.

All access via microprocessor to and from the registers must be 32-bit accesses. There is no support for accesses other than 32-bit. All access via BestComm to and from the registers may be byte, word or longword (32-bit) accesses. However, name based register access is 32-bit aligned.

14.4.1 Top Level Module Memory Map

The FEC implementation requires a 2K Byte memory map space. This is divided into two sections of 512 Bytes and an additional 1KBytes of reserved space. The first 512 Bytes is used for Control and Status Registers. The second contains event/statistic counters held in the MIB block. Table 14-6 defines the top level memory map.

Table 14-6. Module Memory Map

Address	Function
000–1FF	Control/Status Registers
200–3FF	MIB Block Counters, see Table 14-7
400–7FF	Reserved

14.4.2 Control and Status (CSR) Memory Map

TABLE 1. CSR Counters

Address	Mnemonic	Name
000	FEC_ID	FEC_ID register
004	IEVENT	Interrupt Event Register
008	IMASK	Interrupt Enable Register
00C		Reserved
010	R_DES_ACTIVE	Receive Ring Updated Flag
014	X_DES_ACTIVE	Transmit Ring Updated Flag
018-020		Reserved
024	ECNTRL	Ethernet Control Register
028-03C		Reserved
040	MII_DATA	MII Data Register
044	MII_SPEED	MII Speed Register
04C-060		Reserved
064	MIB_CONTROL	MIB Control/Status Register
068-080		Reserved
084	R_CNTRL	Receive Control register
088	R_HASH	Receive Hash
08C-0C0		Reserved
0C4	X_CNTRL	Transmit Control Register
0C8-0E0		Reserved
0E4	PADDR1	Physical Address Low
0E8	PADDR2	Physical Address High+ Type Field
0EC	OP_PAUSE	Opcode + Pause Duration
0F0-114		Reserved
118	IADDR1	Upper 32 bits of individual Hash Table
11C	IADDR2	Lower 32 bits of individual Hash Table

TABLE 1. **CSR Counters**

Address	Mnemonic	Name
120	GADDR1	Upper 32 bits of group Hash Table
124	GADDR2	Lower 32 bits of group Hash Table
128-140		Reserved
144	X_WMRK	Transmit FIFO Watermark
148-180		Reserved
184	RFIFO_DATA	Receive FIFO Data
188	RFIFO_STATUS	Receive FIFO Status
18C	RFIFO_CONTROL	Receive FIFO Control
190	RFIFO_LRF_PTR	Receive FIFO Last Read Frame Pointer
194	RFIFO_LWF_PTR	Receive FIFO Last Write Frame Pointer
198	RFIFO_ALARM	Receive FIFO Alarm Pointer
19C	RFIFO_RDPTR	Receive FIFO Read Pointer
1A0	RFIFO_WRPTR	Receive FIFO Write Pointer
1A4	TFIFO_DATA	Transmit FIFO Data
1A8	TFIFO_STATUS	Transmit FIFO Status
1AC	TFIFO_CONTROL	Transmit FIFO Control
1B0	TFIFO_LRF_PTR	Transmit FIFO Last Read Frame Pointer
1B4	TFIFO_LWF_PTR	Transmit FIFO Last Write Frame Pointer
1B8	TFIFO_ALARM	Transmit FIFO Alarm Pointer
1BC	TFIFO_RDPTR	Transmit FIFO Read Pointer
1C0	TFIFO_WRPTR	Transmit FIFO Write Pointer
1C4	RESET_CNTRL	Reset Control
1C8	XMIT_FSM	Transmit FSM
1CC-1FF		

14.4.3 MIB Block Counters Memory Map

Table 14-7 defines the MIB Counters memory map, which defines the MIB RAM space locations where hardware-maintained counters reside. These fall in the 3200-33FF address range. Counters are divided into two groups.

1. **RMON counters**—are included, which cover ethernet statistics counters defined in RFC 1757. In addition to ethernet statistics group counters, a counter is included to count truncated frames, as FEC only supports frame lengths up to 2047 Bytes. RMON counters are implemented independently for Tx and Rx, to ensure accurate network statistics when operating in full duplex mode.
2. **IEEE counters**—are included, which support the Mandatory and Recommended counter packages defined in Section 5 of ANSI/IEEE Standard 802.3 (1998 edition). FEC supports IEEE Basic Package objects, but does not require MIB block counters. In addition, some recommended package objects supported do not require MIB counters. Counters for Tx and Rx full duplex flow control frames are included.

Table 14-7. MIB Counters

Address	Mnemonic	Description
200	RMON_T_DROP	Count of Frames Not Correctly Counted
204	RMON_T_PACKETS	RMON Tx Packet Count
208	RMON_T_BC_PKT	RMON Tx Broadcast Packets
20C	RMON_T_MC_PKT	RMON Tx Multicast Packets
210	RMON_T_CRC_ALIGN	RMON Tx Packets with CRC/Align error
214	RMON_T_UNDERSIZE	RMON Tx Packets less than 64bytes, good CRC
218	RMON_T_OVERSIZE	RMON Tx Packets greater than MAX_FL bytes, good CRC
21C	RMON_T_FRAG	RMON Tx Packets less than 64bytes, bad CRC
220	RMON_T_JAB	RMONTxPackets greater than MAX_FL bytes, bad CRC
224	RMON_T_COL	RMON Tx collision count
228	RMON_T_P64	RMON Tx 64Byte packets
22C	RMON_T_P65TO127	RMON Tx 65 to 127Byte packets
230	RMON_T_P128TO255	RMON Tx 128 to 255Byte packets
234	RMON_T_P256TO511	RMON Tx 256 to 511Byte packets
238	RMON_T_P512TO1023	RMON Tx 512 to 1023Byte packets
23C	RMON_T_P1024TO2047	RMON Tx 1024 to 2047Byte packets
240	RMON_T_P_GTE2048	RMON Tx packets with greater than 2048Bytes
244	RMON_T_OCTETS	RMON Tx Octets
248	IEEE_T_DROP	Count of Frames Not Counted Correctly
24C	IEEE_T_FRAME_OK	Frames Transmitted OK
250	IEEE_T_1COL	Frames Transmitted with Single Collision
254	IEEE_T_MCOL	Frames Transmitted with Multiple Collisions
258	IEEE_T_DEF	Frames Transmitted after Deferral Delay
25c	IEEE_T_LCOL	Frames Transmitted with Late Collision
260	IEEE_T_EXCOL	Frames Transmitted with Excessive Collisions
264	IEEE_T_MACERR	Frames Transmitted with Tx FIFO Underrun
268	IEEE_T_CSERR	Frames Transmitted with Carrier Sense Error
26C	IEEE_T_SQE	Frames Transmitted with SQE Error
270	T_FDXFC	Flow Control Pause Frames Transmitted
274	IEEE_T_OCTETS_OK	Octet Count for Frames Transmitted w/o Error
278–27C	rsvd	Reserved
280	RMON_R_DROP	Count of frames Not Counted Correctly
284	RMON_R_PACKETS	RMON Rx Packet Count
288	RMON_R_BC_PKT	RMON Rx Broadcast Packets
28C	RMON_R_MC_PKT	RMON Rx Multicast Packets

Table 14-7. MIB Counters (continued)

Address	Mnemonic	Description
290	RMON_R_CRC_ALIGN	RMON Rx Packets with CRC/Align error
294	RMON_R_UNDERSIZE	RMON Rx Packets less than 64Bytes, good CRC
298	RMON_R_OVERSIZE	RMON Rx Packets greater than MAX_FL bytes, good CRC
29C	RMON_R_FRAG	RMON Rx Packets less than 64Bytes, bad CRC
2A0	RMON_R_JAB	RMONRxPackets greater than MAX_FL bytes, bad CRC
2A4	RMON_R_RESVD_0	Reserved
2A8	RMON_R_P64	RMON Rx 64Byte packets
2AC	RMON_R_P65TO127	RMON Rx 65 to 127Byte packets
2B0	RMON_R_P128TO255	RMON Rx 128 to 255Byte packets
2B4	RMON_R_P256TO511	RMON Rx 256 to 511Byte packets
2B8	RMON_R_P512TO1023	RMON Rx 512 to 1023Byte packets
2BC	RMON_R_P1024TO2047	RMON Rx 1024 to 2047Byte packets
2C0	RMON_R_P_GTE2048	RMON Rx packets with greater than 2048Bytes
2C4	RMON_R_OCTETS	RMON Rx Octets
2C8	IEEE_R_DROP	Count of frames not counted correctly
2CC	IEEE_R_FRAME_OK	Frames received OK
2D0	IEEE_R_CRC	Frames received with CRC error
2D4	IEEE_R_ALIGN	Frames received with alignment error
2D8	IEEE_R_MACERR	Rx FIFO overflow count
2DC	R_FDXFC	Flow Control Pause frames received
2E0	IEEE_R_OCTETS_OK	Octet count for frames received without error
2E4–2FC	rsvd	Reserved
300–3FF	rsvd	Reserved

14.5 FEC Registers—MBAR + 0x3000

The FEC uses 37 32-bit registers. These registers are located at an offset from MBAR of 0x3000. Register addresses are relative to this offset. Therefore, the actual register address is **MBAR + 0x3000 + register address**

Hyperlinks to the FEC registers are provided below:

- Section 14-8, *FEC ID Register* (0x3000)
- Section 14-9, *FEC Interrupt Event Register* (0x3004)
- Section 14-10, *FEC Interrupt Enable Register* (0x3008)
- Section 14-11, *FEC Rx Descriptor Active Register* (0x3010)
- Section 14-12, *FEC Tx Descriptor Active Register* (0x3014)
- Section 14-13, *FEC Ethernet Control Register* (0x3024)
- Section 14-14, *FEC MII Management Frame Register* (0x3040)
- Section 14-15, *FEC MII Speed Control Register* (0x3044)
- Section 14-17, *FEC MIB Control Register* (0x3064)
- Section 14-18, *FEC Receive Control Register* (0x3084)
- Section 14-19, *FEC Hash Register* (0x3088)
- Section 14-20, *FEC Tx Control Register* (0x30C4)
- Section 14-21, *FEC Physical Address Low Register* (0x30E4)
- Section 14-22, *FEC Physical Address High Register* (0x30E8)
- Section 14-23, *FEC Opcode/Pause Duration Register* (0x30EC)
- Section 14-24, *FEC Descriptor Individual Address 1 Register* (0x3118)
- Section 14-25, *FEC Descriptor Individual Address 2 Register* (0x311C)
- Section 14-26, *FEC Descriptor Group Address 1 Register* (0x3120)
- Section 14-27, *FEC Descriptor Group Address 2 Register* (0x3124)
- Section 14-28, *FEC Tx FIFO Watermark Register* (0x3144)
- Section 14.7, *FEC Tx FIFO Data Register—MBAR + 0x31A4* (0x3184)
- Section 14.6.1, *FEC Rx FIFO Data Register—MBAR + 0x3184* (0x31A4)
- Section 14-30, *FEC Rx FIFO Status Register* (0x3188)
- Section , *FEC Tx FIFO Status Register* (0x31A8)
- Section 14-31, *FEC Rx FIFO Control Register* (0x318C)
- Section , *FEC Tx FIFO Control Register* (0x31AC)
- Section 14-32, *FEC Rx FIFO Last Read Frame Pointer Register* (0x3190)
- Section , *FEC Tx FIFO Last Read Frame Pointer Register* (0x31B0)
- Section 14-33, *FEC Rx FIFO Last Write Frame Pointer Register* (0x3194)
- Section , *FEC Tx FIFO Last Write Frame Pointer Register* (0x31B4)
- Section 14-34, *FEC Rx FIFO Alarm Pointer Register* (0x3198)
- Section , *FEC Tx FIFO Alarm Pointer Register* (0x31B8)
- Section 14-35, *FEC Rx FIFO Read Pointer Register* (0x319C)
- Section , *FEC Tx FIFO Read Pointer Register* (0x31BC)
- Section 14-36, *FEC Rx FIFO Write Pointer Register* (0x31A0)
- Section , *FEC Tx FIFO Write Pointer Register* (0x31C0)
- Section 14-37, *FEC Reset Control Register* (0x31C4)
- Section 14-38, *FEC Transmit FSM Register* (0x31C8)

14.5.1 FEC ID Register—MBAR + 0x3000

The read-only FEC ID register (FEC_ID) identifies the FEC block and revision.

Table 14-8. FEC ID Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	FEC_ID															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved					DMA	FIFO	Rsvd	FEC_REV							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:15	FEC_ID	Value identifying the FEC 000 = Unique identifier for FEC
16:20	—	Reserved
21	DMA	DMA function is included in the FEC 0 = FEC does not include DMA (BestComm is the DMA engine)
22	FIFO	FIFO function included in the FEC 1 = FEC does include a FIFO
24:31	FEC_REV	Value identifies the FEC revision 00 = Initial revision

14.5.2 FEC Interrupt Event Register—MBAR + 0x3004

When an event occurs that sets a bit in the IEVENT register, an interrupt is generated if the corresponding bit in the interrupt enable register (IMASK) is also set. The IEVENT register bit is cleared if 1 is written to that bit position. A 0 write has no effect. A hardware reset clears this register.

These interrupts can be divided into operational interrupts, transceiver/network error interrupts, and internal error interrupts. Interrupts that may occur in normal operation are:

- GRA
- TFINT
- MII

Interrupts resulting from errors/problems detected in the network or transceiver are:

- HBERR
- BABR
- BABT
- LATE_COL
- COL_RETRY_LIM

Interrupts resulting from internal errors are:

- XFIFO_UN
- XFIFO_ERROR
- RFIFO_ERROR

Some error interrupts are independently counted in the MIB block counters. Software may choose to mask these interrupts, since the errors are visible to network management via the MIB counters.

- HBERR – IEEE_T_SQE
- BABR – RMON_R_OVERSIZE (good CRC), RMON_R_JAB (bad CRC)
- BABT – RMON_T_OVERSIZE (good CRC), RMON_T_JAB (bad CRC)
- LATE_COL – IEEE_T_LCOL
- COL_RETRY_LIM – IEEE_T_EXCOL
- XFIFO_UN – IEEE_T_MACERR

Table 14-9. FEC Interrupt Event Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R W	HBERR	BABR	BABT	GRA	TFINT	Reserved			MII	Rsvd	LATE_COL	COL_ RETRY_LIM	XFIFO_UN	XFIFO_ ERROR	RFIFO_ ERROR	Rsvd
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0	HBERR	Heartbeat Error— interrupt bit indicates HBC is set in the X_CNTRL register and COL input was not asserted within the heartbeat window following a transmission.
1	BABR	Babbling Receive Error—bit indicates frame was received with a length in excess of R_CNTRL.MAX_FL bytes.
2	BABT	Babbling Transmit Error—bit indicates transmitted frame length exceeded R_CNTRL.MAX_FL bytes. This condition is usually caused by a frame that is too long being placed into the transmit data buffer(s). Truncation does not occur.
3	GRA	Graceful Stop Complete—interrupt bit is asserted for one of three reasons. 1 = A graceful stop initiated by setting X_CNTRL.GTS bit is complete. 2 = A graceful stop initiated by setting X_CNTRL.FC_PAUSE bit is complete. 3 = A graceful stop initiated by reception of a valid full duplex flow control "pause" frame is complete. Refer to "Full Duplex Flow Control" section of the Ethernet Operation chapter. A "graceful stop" means the transmitter is put into a pause state after completion of the frame currently being transmitted.
4	TFINT	Transmit frame interrupt. This bit indicates that a frame has been transmitted.
5	—	Reserved
6	—	Reserved
7	—	Reserved
8	MII	MII Interrupt—bit indicates MII completed the data transfer requested.
9	—	Reserved
10	LATE_COL	Bit indicates a collision occurred beyond the collision window (slot time) in half-duplex mode. The frame is truncated with a bad CRC. Remainder of the frame is discarded.
11	COL_RETRY_LIM	Collision Retry Limit—bit indicates a collision occurred on each of 16 successive attempts to transmit the frame. The frame is discarded without being transmitted and transmission of the next frame begins. Only occurs in half-duplex mode.
12	XFIFO_UN	Transmit FIFO Underrun—bit indicates the transmit FIFO became empty before the complete frame was transmitted. A bad CRC is appended to the frame fragment and remainder of frame is discarded.
13	XFIFO_ERROR	Transmit FIFO Error—indicates an error occurred within the forest green version transmit FIFO. When XFIFO_ERROR bit is set, ECNTRL.ETHER_EN is cleared, halting FEC frame processing. When this occurs, software must ensure both the FIFO Controller and BestComm are soft-reset.
14	RFIFO_ERROR	Receive FIFO Error—indicates error occurred within the forest green version RX FIFO. When RFIFO_ERROR bit is set, ECNTRL.ETHER_EN is cleared, halting FEC frame processing. When this occurs, software must ensure both the FIFO Controller and BestComm are soft-reset.
15:31	—	Reserved.

14.5.3 FEC Interrupt Enable Register—MBAR + 0x3008

The IMASK register provides control over the interrupt events allowed to generate an interrupt. All implemented bits in this CSR are R/$\overline{\text{W}}$. This register is cleared by a hardware reset. If corresponding bits in both the IEVENT and IMASK registers are set, the interrupt is signalled to the CPU. The interrupt signal remains asserted until 1 is written to the IEVENT bit (write 1 to clear) or a 0 is written to the IMASK bit.

Table 14-10. FEC Interrupt Enable Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	HBEEN	BREN	BTEN	GRAEN	TFIEN	Reserved			MIIEN	Rsvd	LCEN	CRLEN	XFUNEN	XFERREN	RFERREN	Rsvd
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0	HBEEN	Heartbeat Error Interrupt Enable
1	BREN	Babbling Receiver Interrupt Enable
2	BTEN	Babbling Transmitter Interrupt Enable
3	GRAEN	Graceful Stop Interrupt Enable
4	TFIEN	Transmit Frame Interrupt Enable
5	—	Reserved
6	—	Reserved
7	—	Reserved
8	MIIEN	MII Interrupt Enable
9	—	Reserved
10	LCEN	Late Collision Enable
11	CRLEN	Late Collision Enable
12	XFUNEN	Transmit FIFO Underrun Enable
13	XFERREN	Transmit FIFO Error Enable
14	RFERREN	Receive FIFO Error Enable
15:31	—	Reserved

14.5.4 FEC Rx Descriptor Active Register—MBAR + 0x3010

The FEC descriptor active register is a command register which should be written by the user to indicate that the receive descriptor ring has been updated (empty receive buffers have been produced by the driver with the E bit set).

Whenever the register is written the R_DES_ACTIVE bit is set. This is independent of the data actually written by the user. When set, the FEC will poll the receive descriptor ring and process receive frames (provided ETHER_EN is also set). Once the FEC polls a receive descriptor whose ownership bit is not set, then the FEC will clear the R_DES_ACTIVE bit and cease receive descriptor ring polling until the user sets the bit again, signifying additional descriptors have been placed into the receive descriptor ring.

The R_DES_ACTIVE bit is cleared at reset and by the clearing of ETHER_EN.

Table 14-11. FEC Rx Descriptor Active Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved							R_DES_ACTIVE	Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:6	—	Reserved
7	R_DES_ACTIVE	Set to one when this register is written, regardless of the value written. Cleared by the FEC device whenever no additional “ready” descriptors remain in the receive ring.
8:31	—	Reserved

14.5.5 FEC Tx Descriptor Active Register—MBAR + 0x3014

The FEC descriptor active register is a command register which should be written by the user to indicate that the transmit descriptor ring has been updated (transmit buffers have been produced by the driver with the R bit set in the buffer descriptor).

Whenever the register is written the X_DES_ACTIVE bit is set. This is independent of the data actually written by the user. When set, the FEC will poll the transmit descriptor ring and process transmit frames (provided ETHER_EN is also set). Once the FEC polls a transmit descriptor whose ownership bit is not set, then the FEC will clear the X_DES_ACTIVE bit and cease transmit descriptor ring polling until the sets the bit again, signifying additional descriptors have been placed into the transmit descriptor ring.

The X_DES_ACTIVE bit is cleared at reset and by the clearing of ETHER_EN.

Table 14-12. FEC Tx Descriptor Active Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved							X_DES_ACTIVE	Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:6	—	Reserved
7	X_DES_ACTIVE	Set to one when this register is written, regardless of the value written. Cleared by the FEC device whenever no additional “ready” descriptors remain in the transmit ring.
8:31	—	Reserved

14.5.6 FEC Ethernet Control Register—MBAR + 0x3024

The ECNTRL register is a read/write user register that can enable/disable the FEC. Some fields may be altered by hardware.

Table 14-13. FEC Ethernet Control Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	TAG0	TAG1	TAG2	TAG3	Rsvd	TESTMD	Reserved									
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved													FEC_OE	ETHER_EN	RESET
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:3	TAG[0:3]	This field allows programming and reading the TBUS tag bits. This field is used for debug/test only, and is implemented in two separate 4-bit registers. The “tags_in” register is written to when a sky blue write to this register takes place. This field (tags_in) resets to 1111. During a write cycle to any FEC register other than ECNTRL the tags_in value is driven onto the tbus data bus tag field. During a read cycle the tbus tag field bits is latched and saved in the “tags_out” register. When the ECNTRL register is read the value from “tags_out” shows in the TAG field.
4	—	Reserved
5	TESTMD	Test Mode—used for manufacturing test only. TESTMD resets to 0. This bit forces the bus controller to ignore all bus requests except the one from the SIF.
6:28	—	Reserved
29	FEC_OE	FEC Output Enable—It is a spare bit and has no affect on internal operation.
30	ETHER_EN	Ethernet Enable—When this bit is set, FEC is enabled and Rx/Tx can occur. When bit is cleared, Rx stops immediately; Tx stops after a bad CRC is appended to any frame currently being transmitted. The ETHER_EN bit is altered by hardware under the following conditions: • If ECNTRL.RESET is written to 1 by software, ETHER_EN is cleared. • If error conditions causing the IEVENT.EBERR, XFIFO_ERROR or RFIFO_ERROR bits to set occur ETHER_EN is cleared.
31	RESET	Ethernet Controller Reset—When this bit is set, the equivalent of a hardware reset is done, but it is local to the FEC. ETHER_EN is cleared and all other FEC registers take their reset values. Also, any Tx/Rx currently in progress is abruptly aborted. This bit is automatically cleared by hardware during the reset sequence. The reset sequence takes approximately 8 clock cycles after RESET is written with 1.

14.5.7 FEC MII Management Frame Register—MBAR + 0x3040

This MII_DATA register does not reset to a defined value. The MII_DATA register is used to communicate with the attached MII compatible PHY device(s), providing read/write access to the MII registers.

Writing to the MII_DATA register causes a management frame to be sourced unless the MII_SPEED register has been programmed to 0. When writing to MII_DATA when MII_SPEED = 0, if the MII_SPEED register is then written to a non-zero value, an MII frame is generated with the data previously written to the MII_DATA register. This let MII_DATA and MII_SPEED be programmed in either order if MII_SPEED is currently 0.

Table 14-14. FEC MII Management Frame Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	ST		OP		PA					RA					TA	
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	DATA															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Note: X: Bit does not reset to a defined value.

Bits	Name	Description
0:1	ST	Start of Frame Delimiter—bits must be programmed to 01 for a valid MII management frame.
2:3	OP	Operation Code—field must be programmed to 10 (read) or 01 (write) to generate a valid MII management frame. • A value of 11 causes a “read” frame operation. • A value of 00 causes a “write” frame operation. However, these frames are not MII compliant.
4:8	PA	PHY Address—specifies 1 of up to 32 attached PHY devices.
9:13	RA	Register Address—specifies 1 of up to 32 registers within the specified PHY device.
14:15	TA	TurnAround—must be programmed to 10 to generate a valid MII management frame.
16:31	DATA	Management Frame Data—used for data written to or read from PHY register.

To do a read or write operation, the MII management interface writes to the MII_DATA register. To generate a valid read or write management frame:

- the ST field must be written with a 01
- the OP field must be written with either:
  - 01 (management register write frame), or
  - 10 (management register read frame), and
- the TA field must be written with a 10

If other patterns are written to these fields, a frame is generated, but it does not comply to the IEEE 802.3 MII definition:

- OP field = 1x produces a “read” frame operation, while
- OP field = 0x produces a “write” frame operation.

To generate an IEEE 802.3 compliant MII management interface write frame (write to a PHY register), the user must write the following to the MII_DATA register:

{01 01 PHYAD REGAD 10 DATA}

Writing this pattern causes control logic to shift out the data in the MII_DATA register following a preamble generated by the control state machine. During this time, the MII_DATA register contents are altered as the contents are serially shifted, and is unpredictable if read by the user. When the write management frame operation is complete, the MII_DATAIO_COMPL interrupt is generated. At this time the MII_DATA register contents match the original value written.

To generate an MII Management Interface read frame (read a PHY register) the user must write the following to the MII_DATA register (DATA field content is "don’t care"):

{01 10 PHYAD REGAD 10 XXXX}

Writing this pattern causes control logic to shift out data in the MII_DATA register following a preamble generated by the control state machine. During this time, the MII_DATA register contents are altered as the contents are serially shifted, and is unpredictable if read by the user. When the read management frame operation is complete, the MII_DATAIO_COMPL interrupt is generated. At this time the MII_DATA register contents matches the original value written, except for the DATA field whose contents have been replaced by the value read from the PHY register.

If the MII_DATA register is written while frame generation is in progress, frame contents are altered. Software should use the MII_STATUS register and/or the MII_DATAIO_COMPL interrupt to avoid writing to the MII_DATA register while frame generation is in process.

14.5.8 FEC MII Speed Control Register—MBAR + 0x3044

The MII_SPEED register provides MII clock (MDC pin) frequency control. This allows dropping the MII management frame preamble and provides observability (intended for manufacturing test) of an internal counter used in generating an MDC clock signal.

Table 14-15. FEC MII Speed Control Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved								DIS_PREAMBLE	MII_SPEED						Rsvd
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:23	—	Reserved
24	DIS_PREAMBLE	Asserting this bit causes preamble (32 1s) to not be prepended to the MII management frame. The MII standard allows the preamble to be dropped, if not required by the attached PHY device(s).
25:30	MII_SPEED	Controls the frequency of the MII management interface clock (MDC) relative to ipb_clk. A 0 value in this field “turns off” the MDC and leaves it in low voltage state. Any non-zero value results in the MDC frequency of 1/(MII_SPEED*2) of the ipb_clk frequency. The MII_SPEED field must be programmed with a value to provide an MDC frequency of less than or equal to 2.5 MHz to be compliant with the IEEE MII characteristic. The MII_SPEED must be set to a non-zero value in order to source a read or write management frame. After the management frame is complete, the MII_SPEED register may optionally be set to 0 to turn off the MDC. The MDC generated has a 50% duty cycle except when MII_SPEED is changed during operation (change takes affect following either a rising or falling edge of MDC). If the ipb_clk is 25MHz, programming MII_SPEED field to 0x5 results in a MDC frequency of 25MHz * 1/(5*2) = 2.5 MHz. Table 14-16 shows MII_SPEED optimum values as a function of the ipb_clk frequency.
31	—	Reserved

Table 14-16. Programming Examples for MII_SPEED Register

ipb_clk Frequency	MII_SPEED (Field in Register)	MDC Frequency
25MHz	$5	2.5MHz
33MHz	$7	2.36MHz
40MHz	$8	2.5MHz
50MHz	$A	2.5MHz

14.5.9 FEC MIB Control Register—MBAR + 0x3064

The MIB_CONTROL register is a read/write register used to provide control of and to observe the state of the MIB block. This register is accessed by user software if there is a need to disable the MIB block operation. For example, to clear all MIB counters in RAM the user should disable the MIB block, clear all MIB RAM locations, then enable the MIB block. The MIB_DISABLE bit is reset to 1.

Table 14-17. FEC MIB Control Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	MIB_DISABLE	MIB_IDLE	Reserved													
W																
RESET:	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0	MIB_DISABLE	A read/write control bit. If set, MIB logic halts and MIB counters do not update.
1	MIB_IDLE	A read-only status bit. If set, MIB block is not currently updating MIB counters.
2:31	—	Reserved

14.5.10 FEC Receive Control Register—MBAR + 0x3084

The R_CNTRL register is user programmable. It controls the operational mode of the receive block and should be written only when ETHER_EN = 0 (initialization time).

Table 14-18. FEC Receive Control Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R/W	Reserved					MAX_FL										
RESET:	0	0	0	0	0	1	0	1	1	1	1	0	1	1	1	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R/W	Reserved										FCE	BC_REJ	PROM	MII_MODE	DRT	LOOP
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Name	Description
0:4	—	Reserved
5:15	MAX_FL	Maximum Frame Length—User R/W field. Resets to decimal 1518. The length is measured starting at DA and includes CRC at End Of Frame (EOF). Tx frames longer than MAX_FL causes the BABT interrupt to occur. Rx Frames longer than MAX_FL causes BABR interrupt to occur and sets the EOF buffer descriptor LG bit. The recommended user programmed default value is 1518, or if VLAN Tags are supported, 1522.
16:25	—	Reserved
26	FCE	Flow Control Enable—If asserted, the receiver detects PAUSE frames. On PAUSE frame detection, transmitter stops transmitting data frames for a given duration.
27	BC_REJ	Broadcast Frame Reject—If asserted, frames with DA (destination address) = FFFF_FFFF_FFFF are rejected, unless PROM bit is set. If both BC_REJ and PROM = 1, frames with broadcast DA are accepted and M (MISS) bit is set in the Rx buffer descriptor.
28	PROM	Promiscuous mode—All frames are accepted regardless of address matching.
29	MII_MODE	Selects External Interface Mode—controls the interface mode for Tx/Rx blocks. • Setting bit to 1 selects MII mode. • Setting bit to 0 selects 7wire mode (used only for serial 10Mbps).

Bits	Name	Description
30	DRT	Disable Receive on Transmit 0 = Rx path operates independently of Tx (use for full-duplex or to monitor Tx activity in half-duplex mode). 1 = Disable frames reception while transmitting (normally used for half-duplex mode).
31	LOOP	Internal Loopback—If set, transmitted frames are looped back internal to the device and transmit output signals are not asserted. The system clock is substituted for TX_CLK when LOOP is asserted. DRT must be set to 0 when asserting LOOP.

14.5.11 FEC Hash Register—MBAR + 0x3088

The read-only R_HASH register provides address recognition information from the Rx block about the frame currently being received. These bits provide information used in the address recognition subroutine.

Table 14-19. FEC Hash Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	FCE_DC	MULTI CAST	HASH						Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0	FCE_DC	This is a read-only view of the R_CNTRL register FCE bit.
1	MULTICAST	Set if current Rx frame contained a multi-cast destination address, indicating DA LSB was set. Cleared if current Rx frame does not correspond to a multi-cast address.
2:7	HASH	Corresponds to “hash” value of current Rx frame’s destination address. Hash value is a 6-bit field extracted from least significant portion of CRC register.
8:31	—	Reserved

14.5.12 FEC Tx Control Register—MBAR + 0x30C4

This X_CNTRL register is read/write and is written to configure the transmit block. This register is cleared at system reset. Bits 29:30 should be modified only when ETHER_EN = 0.

Table 14-20. FEC Tx Control Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved											RFC_PAUSE	TFC_PAUSE	FDEN	HBC	GTS
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:26	—	Reserved
27	RFC_PAUSE	This read-only status bit is asserted when a full-duplex flow control pause frame is received. The transmitter is paused for the duration defined in this pause frame. Bit automatically clears when the pause duration is complete.
28	TFC_PAUSE	Assert to transmit a PAUSE frame. When this bit is set, the MAC stops transmission of data frames after the current transmission is complete. At this time, the INTR_EVENT register GRA interrupt is asserted. With transmission of data frames stopped, the MAC transmits a MAC Control PAUSE frame. Next, the MAC clears the TFC_PAUSE bit and resumes transmitting data frames. **Note:** If the transmitter is paused due to user assertion of GTS or reception of a PAUSE frame, MAC may still transmit a MAC Control PAUSE frame.
29	FDEN	Full Duplex Enable—If set, frames are transmitted independent of Carrier Sense and Collision inputs. This bit should only be modified when ETHER_EN is deasserted.
30	HBC	Heartbeat Control—If set, the heartbeat check is done following End Of Transmission (EOT) and the Event Status Register HB bit is set if the collision input does not assert within the heartbeat window. This bit should only be modified when ETHER_EN is deasserted.
31	GTS	Graceful Transmit Stop—When this bit is set, the MAC stops transmission after any frame that is currently being transmitted is complete and the INTR_EVENT register GRA interrupt is asserted. If frame transmission is not currently underway, the GRA interrupt is immediately asserted. Once transmission completes, a “restart” can be done by clearing the GTS bit. The next frame in the transmit FIFO is then transmitted. If an early collision occurs during transmission when GTS = 1, transmission stops after the collision. The frame is transmitted again once GTS is cleared. **Note:** Old frames may exist in the transmit FIFO and be transmitted when GTS is reasserted. To avoid this, deassert ETHER_EN after the GRA interrupt.

14.5.13 FEC Physical Address Low Register—MBAR + 0x30E4

The PADDR1 register is written by the user. This register contains the lower 32bits (Bytes 0,1,2,3) of the 48-bit address used in the address recognition process to compare with the destination address (DA) field of receive frames with an individual DA. In addition, this register is used in Bytes0:3 of the 6-Byte source address field when transmitting PAUSE frames. This register is not reset and must be initialized.

Table 14-21. FEC Physical Address Low Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	PADDR1															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	PADDR1															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Note: X: Bit is not reset and must be initialized.

Bits	Name	Description
0:31	PADDR1	Bytes 0 (bits 31:24), 1 (bits 23:16), 2 (bits 15:8) and 3 (bits 7:0) of the 6-byte individual address used for an exact match, and the Source Address field in PAUSE frames.

14.5.14 FEC Physical Address High Register—MBAR + 0x30E8

The PADDR2 register is written by the user. This register contains the upper 16 bits (bytes 4 and 5) of the 48-bit address used in the address recognition process to compare with the destination address (DA) field of receive frames with an individual DA. In addition, this register is used in Bytes 4 and 5 of the 6-Byte source address field when transmitting PAUSE frames. Bits 16:31 of XMIT**.**PADDR2 contain a constant type field (hex 8808) used for transmission of PAUSE frames. This register is not reset and bits 0:15 must be initialized.

Table 14-22. FEC Physical Address High Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	PADDR2															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	TYPE															
W																
RESET:	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0

Note: X: Bit is not reset and must be initialized.

Bits	Name	Description
0:15	PADDR2	Bytes 4 (bits 31:24) and 5 (bits 23:16) of the 6-byte individual address used for an exact match, and the Source Address field in PAUSE frames.
16:31	TYPE	These 16 bits are a constant value, hex 8808.

14.5.15 FEC Opcode/Pause Duration Register—MBAR + 0x30EC

The OP_PAUSE register is read/write accessible. This register contains the 16-bit opcode, and 16-bit pause duration fields used in transmission of a PAUSE frame. The opcode field is a constant value, hex 0001. When another node detects a PAUSE frame, that node pauses transmission for the duration specified in the pause duration field. This register is not reset and bits 16:31 must be initialized.

Table 14-23. FEC Opcode/Pause Duration Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	OPCODE															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	PAUSE_DUR															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Note: X: Bit is not reset and must be initialized.

Bits	Name	Description
0:15	OPCODE	Opcode field used in PAUSE frames. Bits are a constant value, hex 0001.
16:31	PAUSE_DUR	Pause Duration field used in PAUSE frames.

14.5.16 FEC Descriptor Individual Address 1 Registe—MBAR + 0x3118

The IADDR1 register is written by the user. This register contains the upper 32 bits of the 64-bit individual address hash table used in the address recognition process to check for possible match with the DA field of receive frames with an individual DA. This register is not reset and must be initialized.

Table 14-24. FEC Descriptor Individual Address 1 Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	IADDR1															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	IADDR1															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Note: X: Bit is not reset and must be initialized.

Bits	Name	Description
0:31	IADDR1	The upper 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a unicast address. • Bit 31 contains hash index bit 63. • Bit 0 contains hash index bit 32.

14.5.17 FEC Descriptor Individual Address 2 Register—MBAR + 0x311C

The IADDR2 register is written by the user. This register contains the lower 32 bits of the 64-bit individual address hash table used in the address recognition process to check for possible match with the DA field of receive frames with an individual DA. This register is not reset and must be initialized.

Table 14-25. FEC Descriptor Individual Address 2 Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	IADDR2															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	IADDR2															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Note: X: Bit is not reset and must be initialized.

Bits	Name	Description
0:31	IADDR2	The lower 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a unicast address. • Bit 31 contains hash index bit 31. • Bit 0 contains hash index bit 0.

14.5.18 FEC Descriptor Group Address 1 Register—MBAR + 0x3120

The GADDR1 register is written by the user. This register contains the upper 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. This register must be initialized.

Table 14-26. FEC Descriptor Group Address 1 Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	GADDR1															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	GADDR1															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Note: X: Bit is not reset and must be initialized.

Bits	Name	Description
0:31	GADDR1	The GADDR1 register contains the upper 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. • Bit 31 contains hash index bit 63. • Bit 0 contains hash index bit 32.

14.5.19 FEC Descriptor Group Address 2 Register—MBAR + 0x3124

The GADDR2 register is written by the user. The GADDR2 register contains the lower 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. This register must be initialized.

Table 14-27. FEC Descriptor Group Address 2 Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	GADDR2															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	GADDR2															
W																
RESET:	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Note: X: Bit is not reset and must be initialized.

Bits	Name	Description
0:31	GADDR2	The GADDR2 register contains the lower 32 bits of the 64-bit hash table used in the address recognition process for receive frames with a multicast address. • Bit 31 contains hash index bit 31. • Bit 0 contains hash index bit 0.

14.5.20 FEC Tx FIFO Watermark Register—MBAR + 0x3144

The X_WMRK register is a user programmable 4-bit read/write register that controls the amount of data required in the transmit FIFO before transmission of a frame can begin. This lets the user minimize transmit latency (X_WMRK = 0000) or allows for larger bus access latency (X_WMRK = 1111) due to contention for the system bus. Setting the watermark to a high value minimizes the risk of transmit FIFO underrun due to contention for the system bus. The X_WMRK register resets to 0.

NOTE

This register value may need to be customized by software for specific FEC applications to be compatible with specific FIFO/system bus access latency requirements.

Table 14-28. FEC Tx FIFO Watermark Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved												X_WMRK			
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:28	—	Reserved
28:31	X_WMRK	Transmit FIFO Watermark—Frame transmission begins: • If the number of bytes selected by this field are written into the transmit FIFO, or • if an EOF is written to the FIFO, or • if the FIFO is full before the selected number of bytes are written. Options are: 0000 = 64 Bytes written to FIFO. 0001 = 128 Bytes written to FIFO. 0010 = 192 Bytes written to FIFO. 0011 = 256 Bytes written to FIFO. 0100 = 320 Bytes written to FIFO. 0101 = 384 Bytes written to FIFO. 0110 = 448 Bytes written to FIFO. 0111 = 512 Bytes written to FIFO. 1000 = 576 Bytes written to FIFO. 1001 = 640 Bytes written to FIFO. 1010 = 704 Bytes written to FIFO. 1011 = 768 Bytes written to FIFO. 1100 = 832 Bytes written to FIFO. 1101 = 896Bytes written to FIFO. 1110 = 960 Bytes written to FIFO. 1111 = 1024 Bytes written to FIFO.

14.6 FIFO Interface

The programming interface to the FIFO allows access to Data, Status, Control, Last Write Pointer, Last Read Pointer, Alarm, Read and Write Pointers for Transmit and Receive configurations. The FIFO can be accessed by byte, word, or longword, but all accesses must be aligned with the most significant byte (big endian) of the data port. BestComm supports byte, word or longword accesses. The processor supports longword access only. All register name access is longword aligned

Table 14-29. FIFO Interface Register Map

Address	byte0	byte1	byte2	byte3	Description
0x184	Data	Data	Data	Data	Receive FIFO Data
0x188	Stat	Stat			Receive FIFO Status
0x18C	Ctl				Receive FIFO Control
0x190			LRF	LRF	Receive Last Read Frame Pointer
0x194			LWF	LWF	Receive Last Write Frame Pointer
0x198			Alarm	Alarm	Receive (High/Low) Alarm Pointer
0x19C			Read	Read	Receive FIFO Read Pointer
0x1A0			Write	Write	Receive FIFO Write Pointer
0x1A4	Data	Data	Data	Data	Transmit FIFO Data
0x1A8	Stat	Stat			Transmit FIFO Status
0x1AC	Ctl				Transmit FIFO Control
0x1B0			LRF	LRF	Transmit Last Read Frame Pointer

Table 14-29. FIFO Interface Register Map (continued)

Address	byte0	byte1	byte2	byte3	Description
0x1B4			LWF	LWF	Transmit Last Write Frame Pointer
0x1B8			Alarm	Alarm	Transmit (High/Low) Alarm Pointer
0x1BC			Read	Read	Transmit FIFO Read Pointer
0x1C0			Write	Write	Transmit FIFO Write Pointer

14.6.1 FEC Rx FIFO Data Register—MBAR + 0x3184

14.7 FEC Tx FIFO Data Register—MBAR + 0x31A4

The RFIFO_DATA and TFIFO_DATA registers are the main interface port for the Transmit and Receive FIFO. Data which is to be buffered in the FIFO, or has been buffered in the FIFO, is accessed through this register.

14.7.1 FEC Rx FIFO Status Register—MBAR + 0x3188

14.8 FEC Tx FIFO Status Register—MBAR + 0x31A8

The RFIFO_STATUS and TFIFO_STATUS registers contain bits which provide information about the status of the FIFO controller. The bits marked sticky are cleared by writing a ‘1’ to their positions.

Table 14-30. FEC Rx FIFO Status Register
FEC Tx FIFO Status Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved				Frame[0:3]				Rsvd	Error	UF	OF	FR	Full	Alarm	Empty
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Name	Description
0:3	—	Reserved
4:7	Frame[0:3]	Frame Indicator – READ ONLY This bus provides a frame status indicator for non-DMA applications. Frame[0] = A frame boundary has occurred on the [31:24] byte of the data bus. Frame[1] = A frame boundary has occurred on the [23:16] byte of the data bus. Frame[2] = A frame boundary has occurred on the [15:8] byte of the data bus. Frame[3] = A frame boundary has occurred on the [7:0] byte of the data bus.
8	---	Reserved
9	Error	FIFO Error – Sticky, Write To Clear. This bit signifies that an error has occurred in the FIFO controller. Errors can be caused by underflow, overflow,or pointers being out of bounds. This bit will remain set until this bit of the FIFO status register has been written with a 1.

Bits	Name	Description
10	UF	UF FIFO Underflow – Sticky, Write To Clear This bit signifies the read pointer has surpassed the write pointer. This bit will remain set until this bit of the FIFO status register has been written with a 1.
11	OF	OF FIFO Overflow – Sticky, Write To Clear This bit signifies the write pointer has surpassed the read pointer. This bit will remain set until this bit of the FIFO status register has been written with a 1.
12	FR	FR Frame Ready – Read Only The FIFO has requested attention because there is framed data ready. All complete frames must be read from the FIFO to clear this alarm. This alarm will only be asserted while in frame mode.
13	Full	Full Alarm – Read Only The FIFO has requested attention because it is full. The FIFO must be read to clear this alarm.
14	Alarm	FIFO Alarm – Read Only The FIFO has requested attention because it has determined an alarm condition. The specific alarm condition detected is dependent upon the FIFO direction (Transmit or Receive); if it is a Transmit FIFO, then the FIFO alarm output pin provides indication of a low level, asserting when there is less than alarm bytes of data remaining in the FIFO, and deasserting when there are less than 4* granularity free bytes remaining. When the FIFO is configured to Receive, the FIFO alarm provides high level indication, asserting when there are less than alarm bytes free in the FIFO, and deasserting when there are less than granularity bytes of data remaining. This signal can be cleared by reading or writing (as appropriate) the FIFO, or manipulating the FIFO pointers.
15	Empty	Empty – Read Only The FIFO has requested attention because it is empty. The FIFO must be written to clear this alarm.
16:31	---	Reserved

14.8.1 FEC Rx FIFO Control Register—MBAR + 0x318C
FEC Tx FIFO Control Register—MBAR + 0x31AC

The RFIFO_CONTROL and TFIFO_CONTROL registers provide programmability of many FIFO behaviors, from last transfer granularity to frame operation. Last transfer granularity allows the user to control when the FIFO controller stops requesting data transfers through the FIFO alarm. When the alarm is configured as a Receive FIFO, the granularity value is the GR[2:0] value. When the alarm is configured as a Transmit FIFO, the granularity value is four times the GR[2:0] value, or the pipeline depth. The frame bit of the control register provides a capability to enable and control the FIFO controller's ability to view data on a packetized basis. The FIFO controller also has the programmable capability to not request attention after it has received a complete frame until Ethernet has reported completion of transmission. Frame mode supersedes the FIFO granularity bits, through the assertion of a hardware signal to BestComm.

Table 14-31. FEC Rx FIFO Control Register
FEC Tx FIFO Control Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Rsvd	WFR[1:0]		COMP	FRAME	GR[2:0]			Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0	—	Reserved
1:2	WFR[1:0]	Write Frame 01 = the FIFO controller assumes the next write to its data port is the next to last write. 10 = the FIFO controller assumes the next write to its data port is status / control information.
3	COMP	COMP Re-enable Requests on Frame Transmission Completion. When this bit is set, the FIFO controller will not request attention between receiving the last data of the frame from the BestComm until the peripheral acknowledges transmission of the frame.
4	FRAME	Frame Mode Enable. When this bit is set, the FIFO controller monitors frame done information from the peripheral or BestComm. Setting this bit also enables the other frame control bits in this register, as well as other frame functions. This bit must be set to use frame functions.
5:7	GR[2:0]	Last Transfer Granularity. These bits define the deassertion point for the “high” service request and also define the deassertion point for the “low” service request. A “high” service request is deasserted when there are less than GR[2:0] data bytes remaining in the FIFO. A “low” service request is deasserted when there are less than (4 * GR[2:0]) free bytes remaining in the FIFO.

14.8.2 FEC Rx FIFO Last Read Frame Pointer Register—MBAR + 0x3190 FEC Tx FIFO Last Read Frame Pointer Register—MBAR + 0x31B0

The RFIFO_LRF_PTR and TFIFO_LRF_PTR are a FIFO-maintained pointer which indicates the location of the start of the most recently read frame, or the start of the frame currently in transmission. The LRFP updates on FIFO read data accesses to a frame boundary. The LRFP can be read and written for debug purposes. For the frame retransmit function, the LRFP indicates which point to begin retransmission of the data frame. The LRFP carries validity information, however, there are no safeguards to prevent retransmitting data which has been overwritten. When FRAME is not set, then this pointer has no meaning.

Table 14-32. FEC Rx FIFO Last Read Frame Pointer Register
FEC Tx FIFO Last Read Frame Pointer Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved						LRFP[9:0]									
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:21	—	Reserved
22:31	LRFP[9:0]	LRFP Last Read Frame Pointer. This pointer indicates the start of the last data frame read from the FIFO by the peripheral.

14.8.3 FEC Rx FIFO Last Write Frame Pointer Register—MBAR + 0x3194 FEC Tx FIFO Last Write Frame Pointer Register—MBAR + 0x31B4

The RFIFO_LWF_PTR and TFIFO_LWF_PTR are a FIFO maintained pointer which indicates the location of the start of the last frame written into the FIFO. The LWFP updates on FIFO write data accesses which create a frame boundary, whether that be by setting the writeFrameCtl control bit, or by feeding a frame bit in on the appropriate bus. The LWFP can be read and written for debug purposes. For the frame discard function, the LWFP divides the valid data region of the FIFO (the area in-between the read and write pointers) into framed and unframed data. Data between the LWFP and write pointer constitutes an incomplete frame, while data between the read pointer and the LWFP has been received as whole frames. When FRAME is not set, then this pointer has no meaning.

Table 14-33. FEC Rx FIFO Last Write Frame Pointer Register FEC Tx FIFO Last Write Frame Pointer Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved						LRFP[9:0]									
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:21	—	Reserved
22:31	LWFP[9:0]	LRFP Last WriteFrame Pointer. This pointer indicates the start of the last data frame written into the FIFO by the peripheral.

14.8.4 FEC Rx FIFO Alarm Pointer Register—MBAR + 0x3198 FEC Tx FIFO Alarm Pointer Register—MBAR + 0x31B8

RFIFO_ALARM and TFIFO_ALARM include pointer which provide high/low level alarm information to the user integration logic and the BestComm interface. A low level alarm reports lack of data; a high level alarm reports lack of space. The alarm pointer is interpreted depending on the state of the FIFO transmit input pin: if FIFO transmit = ‘1’, then the alarm is represented in terms of data bytes, if FIFO Transmit = ‘0’, the alarm is represented in terms of free bytes. This programmable alarm can warn the system when the FIFO is almost full of data (FIFO Transmit = ‘0’), or when the FIFO is almost out of data (FIFO Transmit = ‘1’). This register is programmed to the upper limit for the number of bytes in the FIFO of data, when FIFO transmit is negated, or space, when FIFO transmit is asserted, before an internal alarm is set. Any time the amount of data or space in the FIFO is above the indicated amount, the alarm will be set. The alarm is cleared when there is less data or space than is defined as the FIFO granularity or pipeline depth. The number of bits in the alarm pointer register will vary with the address space of the FIFO memory, and the alarm pointer is initialized to zero.

**Table 14-34. FEC Rx FIFO Alarm Pointer Register
FEC Tx FIFO Alarm Pointer Register**

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved						Alarm[9:0]									
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:21	—	Reserved
22:31	Alarm[9:0]	Alarm Pointer. This pointer indicates the point at (or below) which to assert the FIFO alarm signal. This value is compared with data or free bytes, depending upon the state of FIFO Transmit (FIFO Transmit = ‘1’, alarm measures data bytes).

14.8.5 FEC Rx FIFO Read Pointer Register—MBAR + 0x319C FEC Tx FIFO Read Pointer Register—MBAR + 0x31BC

The RFIFO_RDPTR and TFIFO_RDPTR are a FIFO-maintained pointer which point to the next FIFO location to be read. The read pointer can be both read and written.

**Table 14-35. FEC Rx FIFO Read Pointer Register
FEC Tx FIFO Read Pointer Register**

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved						READ[9:0]									
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:21	—	Reserved
22:31	READ[9:0]	Read Pointer. This pointer indicates the next location to be read by the FIFO controller.

14.8.6 FEC Rx FIFO Write Pointer Register—MBAR + 0x31A0 FEC Tx FIFO Writer Pointer Register—MBAR + 0x31C0

The RFIFO_WRPTR and TFIFO_WRPTR are a FIFO-maintained pointer which point to the next FIFO location to be written. The write pointer can be both read and written.

Table 14-36. FEC Rx FIFO Write Pointer Register FEC Tx FIFO Write Pointer Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved						WRITE[9:0]									
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:21	—	Reserved
22:31	WRITE[9:0]	WRITE Pointer. This pointer indicates the next location to be written by the FIFO controller.

14.8.7 FEC Reset Control Register—MBAR + 0x31C4

The RESET_CNTRL register allows reset of the FIFO controllers.

Table 14-37. FEC Reset Control Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved						RCTL[1]	RCTL[0]	Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 1-1.

Bits	Name	Description
0:5	—	Reserved
6	RCTL[1]	0 = Do not Reset FIFO controllers. 1 = Reset FIFO controllers.

Table 1-1.

Bits	Name	Description
7	RCTL[0]	0 = Disable fec_enable as a reset to FIFO controllers. 1 = Enable fec_enable as a reset to FIFO controllers.
8:31	---	Reserved

14.8.8 FEC Transmit FSM Register—MBAR + 0x31C8

The transmit finite state machine register (XMIT_FSM) controls operation of appending CRC. Typical use is enabled and CRC appended.

Table 14-38. FEC Transmit FSM Register

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
R	Reserved						XFSM[1]	XFSM[0]	Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31 lsb
R	Reserved															
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
0:5	—	Reserved
6	XFSM[1]	0 = Do not append CRC. 1 = Append CRC (typical use).
7	XFSM[0]	0 = Disable CRC FSM. 1 = Enable CRC FSM (typical use is enabled).
8:31	---	Reserved

14.9 Initialization Sequence

This section describes which registers are hardware reset, which are reset by the FEC, and what locations the user must initialize prior to enabling the FEC.

14.9.1 Hardware Controlled Initialization

Some registers in the FEC are reset by internal logic. Specifically those registers are control logic that generate interrupts, cause outputs to be asserted and in general, configuration control bits.

Other registers are reset when the ETHER_EN bit is not asserted (i.e., cleared). To halt operation ETHER_EN is deasserted by either a hard reset or by software. By deasserting ETHER_EN configuration control registers such as X_CNTRL and R_CNTRL are not reset, but the entire data path is reset.

Table 14-39 shows the effect deasserting ETHER_EN has on Ethernet MAC operation and registers.

Table 14-39. ETHER_EN De-Assertion Affect on FEC

Register/Machine	Reset Value
XMIT block	Transmission Aborted (bad CRC appended)
RECV block	Receive activity aborted
Tx/Rx FIFO	Reset control logic dependent on reset_cntrl

14.9.2 User Initialization (Prior to Asserting ETHER_EN)

The user needs to initialize portions of the FEC prior to setting the ETHER_EN bit. The exact values depend on the particular application; the sequence of writing the registers is not important. Ethernet MAC registers requiring initialization are defined in Table 14-40.

Table 14-40. User Initialization (Before ETHER_EN)

Description
Initialize IMASK
Clear IEVENT (write FFFF_FFFF)
X_WMRK (optional)
IADDR2/IADDR1
GADDR1/GADDR2
PADDR1/PADDR2
OP_PAUSE (only needed for FDX flow control)
R_CNTRL
X_CNTRL
MII_SPEED (optional)
Clear MIB_RAM (locations 200–2FC)

14.9.2.1 Microcontroller Initialization

In the FEC the descriptor control RISC initializes some registers after ETHER_EN is asserted. After the Microcontroller initialization sequence is complete, hardware is ready for operation.

Table 14-41 shows RISC initialization operations common to the FEC.

Table 14-41. Microcontroller Initialization (FEC)

Description
Initialize BackOff random number seed
Activate Receiver
Activate Transmit

14.9.3 Frame Control/Status Words

In the FEC transmit frame control words and receive frame status words cross the following the end of frame data. These words are marked with a type value of 10 and have the following formats.

14.9.3.1 Receive Frame Status Word

Table 14-2 below defines the format for the receive frame status word.

Table 14-42. Receive Frame Status Word Format

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
0	0	0	0	1 (Last)	0	0	M	BC	MC	LG	NO	0	CR	OV	TR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	FRAME_LENGTH										

Bits 31-28, 26-25, 19 and 15-11—Reserved

L—Last in Frame, written by the FEC

The buffer is not the last in a frame.

The buffer is the last in a frame.

BC—Will be set if the DA is broadcast (FF-FF-FF-FF-FF-FF)

MC—Will be set if the DA is multicast and not BC

LG—Rx Frame Length Violation, written by the FEC.

A frame length greater than R_CNTRL.MAX_FL was recognized. This bit is valid only if the L-bit is set. The receive data is not altered in any way unless the length exceeds 2047 bytes.

NO—Rx Non-octet Aligned Frame, written by the FEC.

A frame that contained a number of bits not divisible by 8 was received, and the CRC check that occurred at the preceding byte boundary generated an error. This bit is valid only if the L-bit is set. If this bit is set the CR bit will not be set.

CR—Rx CRC Error, written by the FEC.

This frame contains a CRC error and is an integral number of octets in length. This bit is valid only if the L-bit is set.

OV—Overrun, written by the FEC.

A receive FIFO overrun occurred during frame reception. If this bit is set, the other status bits, M, LG, NO, SH, CR, and CL lose their normal meaning and will be zero. This bit is valid only if the L-bit is set.

TR—Rx Frame Truncated

Will be set if the receive frame is truncated (frame length > 2047 bytes). If the TR bit is set the frame should be discarded and the other error bits should be ignored as they may be incorrect.

FRAME_LENGTH— Length of Received Frame

14.9.3.2 Transmit Frame Control Word

The only requirement for this control word is to have the TC and ABC bits valid. The TC bit defines whether the transmit block should append the CRC (TC = 1) or not (TC = 0) for the current frame. The ABC bit defines whether the transmit block should append a bad CRC (ABC = 1), independent of the TC value. Refer to Table 14-43 below for the format of the transmit frame control word.

Table 14-43. Transmit Frame Control Word Format

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
					TC	ABC									

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Bits 31-27, 24-0—Reserved

TC—Transmit CRC, written by user

0 = End transmission immediately after the last data byte.

1 = Transmit the CRC sequence after the last data byte.

ABC—Append Bad CRC, written by user

0 = No affect

1 = Transmit the CRC sequence inverted after the last data bye (regardless of TC value).

14.9.4 Network Interface Options

The FEC supports both an MII interface for 10/100 Mbps Ethernet and a 7-wire serial interface for 10 Mbps Ethernet. The interface mode is selected by the MII_MODE bit in the R_CNTRL register. In MII mode (R_CNTRL.MII_MODE = 1), there are 18 signals defined by the 802.3 standard and supported by the FEC. These are shown in Table 14-1:

The 7-Wire serial interface (R_CNTRL.MII_MODE = 0) operates in what is generally referred to as the “AMD” mode. FEC Frame Transmission

The Ethernet transmitter is designed to work with almost no intervention from software. Once ETHER_EN is asserted and data appears in the transmit FIFO the Ethernet MAC is able to transmit onto the network.

When the transmit FIFO fills to the watermark (defined by the X_WMRK register), the MAC transmit logic will assert TX_EN and start transmitting the preamble sequence, the start frame delimiter, and then the frame information from the FIFO. However, the controller defers the transmission if the network is busy (carrier sense is asserted). Before transmitting, the controller waits for carrier sense to become inactive, then determines if carrier sense stays inactive for 60 bit times. If so, then the transmission begins after waiting an additional 36 bit times (96 bit times after carrier sense originally became inactive).

If a collision occurs during transmission of the frame (half-duplex mode), the ethernet controller follows the specified backoff procedures and attempts to retransmit the frame until the retry limit is reached. The transmit FIFO stores at least the first 64 bytes of the transmit frame, so that they do not have to be retrieved from system memory in case of a collision. This improves bus utilization and latency in case immediate retransmission is necessary.

When all the frame data has been transmitted, the FCS (32-bit CRC) bytes are appended if the TC bit is set in the transmit frame control word. If the ABC bit is set in the transmit frame control word, a bad CRC will be appended to the frame data regardless of the TC bit value. Following the transmission of the CRC, the ethernet controller writes the frame status information to the MIB block. Short frames are automatically padded by the transmit logic (if the TC bit in the transmit buffer descriptor for the end of frame buffer = 1).

The FEC frame interrupts may be generated as determined by the settings in the IMASK register.

Transmit error interrupts are HBERR, BABT, LATE_COL, COL_RETRY_LIM, XFIFO_UN and XFIFO_ERROR. If the transmit frame length exceeds MAX_FL bytes the BABT interrupt will be asserted, however the entire frame will be transmitted (no truncation).

To pause transmission, set the GTS (Graceful Transmit Stop) bit in the X_CNTRL register. When the GTS is set the FEC transmitter stops immediately if transmission is not in progress; otherwise, it continues transmission until the current frame either finishes or terminates with a collision. After the transmitter has stopped the GRA (Graceful Stop Complete) interrupt is asserted. If GTS is cleared, the FEC resumes transmission with the next frame.

The ethernet controller transmits bytes least significant bit first.

14.9.5 FEC Frame Reception

The FEC receiver is designed to work with almost no intervention from the host and can perform address recognition, CRC checking, short frame checking and maximum frame length checking.

When the driver enables the FEC receiver by asserting ETHER_EN it will immediately start processing receive frames. When RX_DV asserts, the receiver will first check for a valid PA/SFD header. If the PA/SFD is valid it will be stripped and the frame will be processed by the receiver. If a valid PA/SFD is not found the frame will be ignored.

In 7-wire serial mode, the first 16 bit times of RX_D0 following assertion of RX_DV (RENA) are ignored. Following the first 16 bit times the data sequence is checked for alternating 1s and 0s. If a 11 or 00 data sequence is detected during bit times 17 to 21, the remainder of the frame is ignored. After bit time 21, the data sequence is monitored for a valid SFD (11). If a 00 is detected, the frame is rejected. When a 11 is detected, the PA/SFD sequence is complete.

In MII mode the receiver checks for at least one byte matching the SFD. Zero or more PA bytes may occur, but if a 00 bit sequence is detected prior to the SFD byte, the frame is ignored.

After the first 6 bytes of the frame have been received, the FEC performs address recognition on the frame.

Once a collision window (64 bytes) of data has been received and if address recognition has not rejected the frame, the receive FIFO is signalled that the frame is "accepted" and may be passed on to the DMA. If the frame is a runt (due to collision) or is rejected by address recognition, the receive FIFO is notified to "reject" the frame. Thus, no collision fragments are presented to the user except late collisions, which indicate serious LAN problems.

During reception, the ethernet controller checks for various error conditions and once the entire frame is written into the FIFO, a 32-bit frame status word is written into the FIFO. This status word contains the M, BC, MC, LG, NO, SH, CR, OV and TR status bits, and the frame length.

The ethernet controller receives serial data LSB first.

14.9.6 Ethernet Address Recognition

The FEC filters the received frames based on destination address (DA) type — individual (unicast), group (multicast) or broadcast (all-ones group address). The difference between an individual address and a group address is determined by the I/G bit in the destination address field. A flowchart for address recognition on received frames is illustrated in the figures below.

Address recognition is accomplished through the use of the receive block and microcode running on the microcontroller. The flowchart shown in Figure14-4 illustrates the address recognition decisions made by the receive block, while Figure 14-5 illustrates the decisions made by the microcontroller.

If the DA is a broadcast address and broadcast reject (R_CNTRL.BC_REJ) is deasserted, then the frame will be accepted unconditionally as shown in Figure 14-4. Otherwise, if the DA in not a broadcast address the microcontroller runs the address recognition subroutine as shown in Figure 14-5.

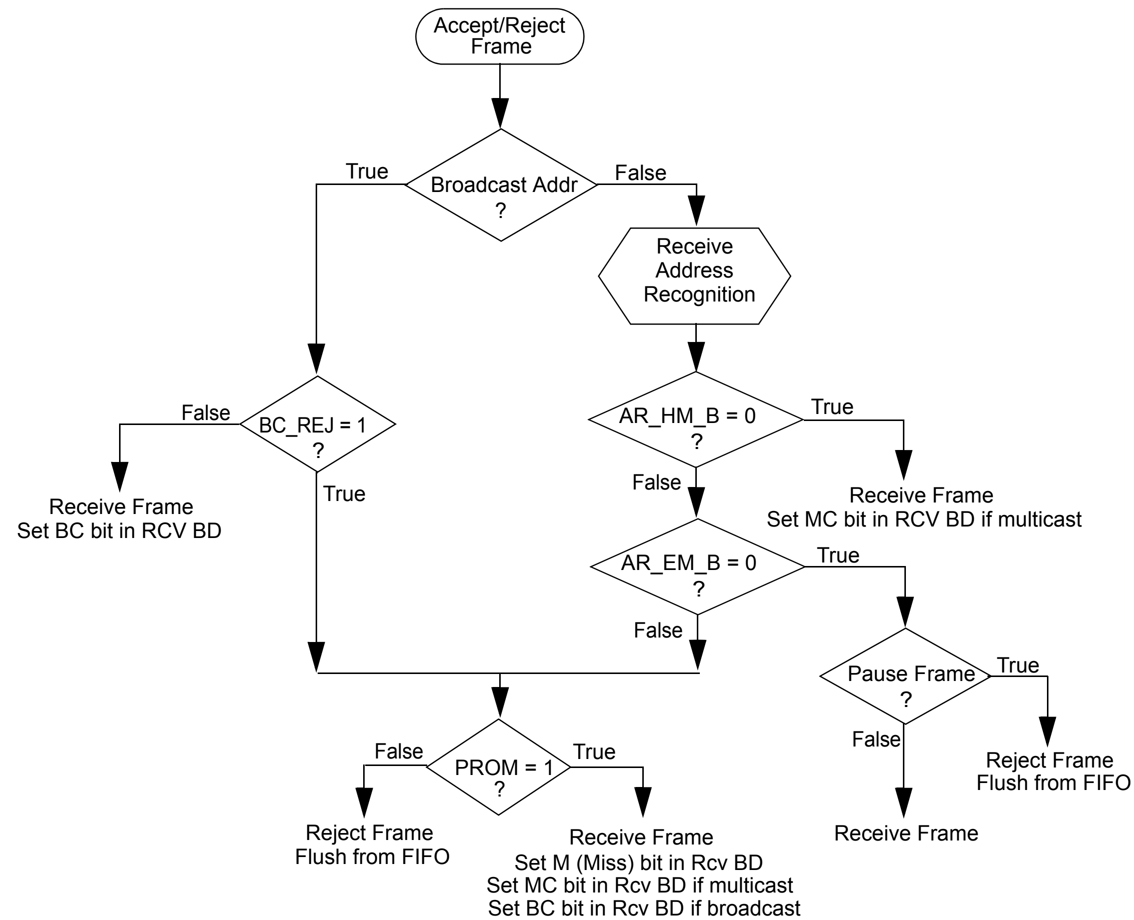
If the DA is a group (multicast) address and flow control is disabled the microcontroller will perform a group hash table lookup using the 64-entry hash table programmed in GADDR1 and GADDR2. If a hash match occurs AR_HM_B (address recognition hash match bar) is set to 0 and the receiver accepts the frame. If flow control is enabled the microcontroller will do an exact address match check between the DA and the designated PAUSE DA in registers XMIT.FDXFC_DA1 and XMIT.FDXFC_DA2. In the case where a PAUSE DA exact match occurs AR_EM_B (address recognition exact match bar) is set to 0. If the receive block determines that the received frame is a valid PAUSE frame the frame will be rejected. Note the receiver will detect a PAUSE frame with the DA field set to either the designated PAUSE DA or the unicast physical address.

If the DA is the individual (unicast) address the microcontroller performs an individual exact match comparison between the DA and 48-bit physical address that the user programs in the PADDR1 and PADDR2 registers. If an exact match occurs AR_EM_B is set to 0; otherwise, the microcontroller does an individual hash table lookup using the 64-entry hash table programmed in registers IADDR1 and IADDR2. In the case of an individual hash match AR_HM_B is set to 0. Again, the receiver will accept or reject the frame based on PAUSE frame detection, shown in Figure 14-4.

If neither a hash match (group or individual) nor an exact match (group or individual) occur both AR_HM_B and AR_EM_B are set to 1. In this case, if promiscuous mode is enabled (R_CNTRL.PROM = 1), then the frame will be accepted and the MISS bit in the receive buffer descriptor is set; otherwise, the frame will be rejected and the MISS bit will be cleared.

Similarly, if the DA is a broadcast address, broadcast reject (R_CNTRL.BC_REJ) is asserted and promiscuous mode is enabled. Then the frame will be accepted and the MISS bit in the receive buffer descriptor is set; otherwise, the frame will be rejected and the MISS bit will be cleared.

In general, when a frame is rejected it is flushed from the FIFO.



NOTES:
BC_REJ - field in R_CNTRL register (BroadCast REJect)
Check Address - microcode Address Recognition subroutine; returns AR_HM_B and AR_EM_B
AR_EM_B - bit in RECV.AR_DONE register (address recognition exact match bar)
AR_HM_B - bit in RECV.AR_DONE register (address recognition hash match bar)
PROM - field in R_CNTRL register (PROMiscous mode)
Pause Frame - valid PAUSE frame received

Figure 14-2. Ethernet Address Recognition - receive block decisions

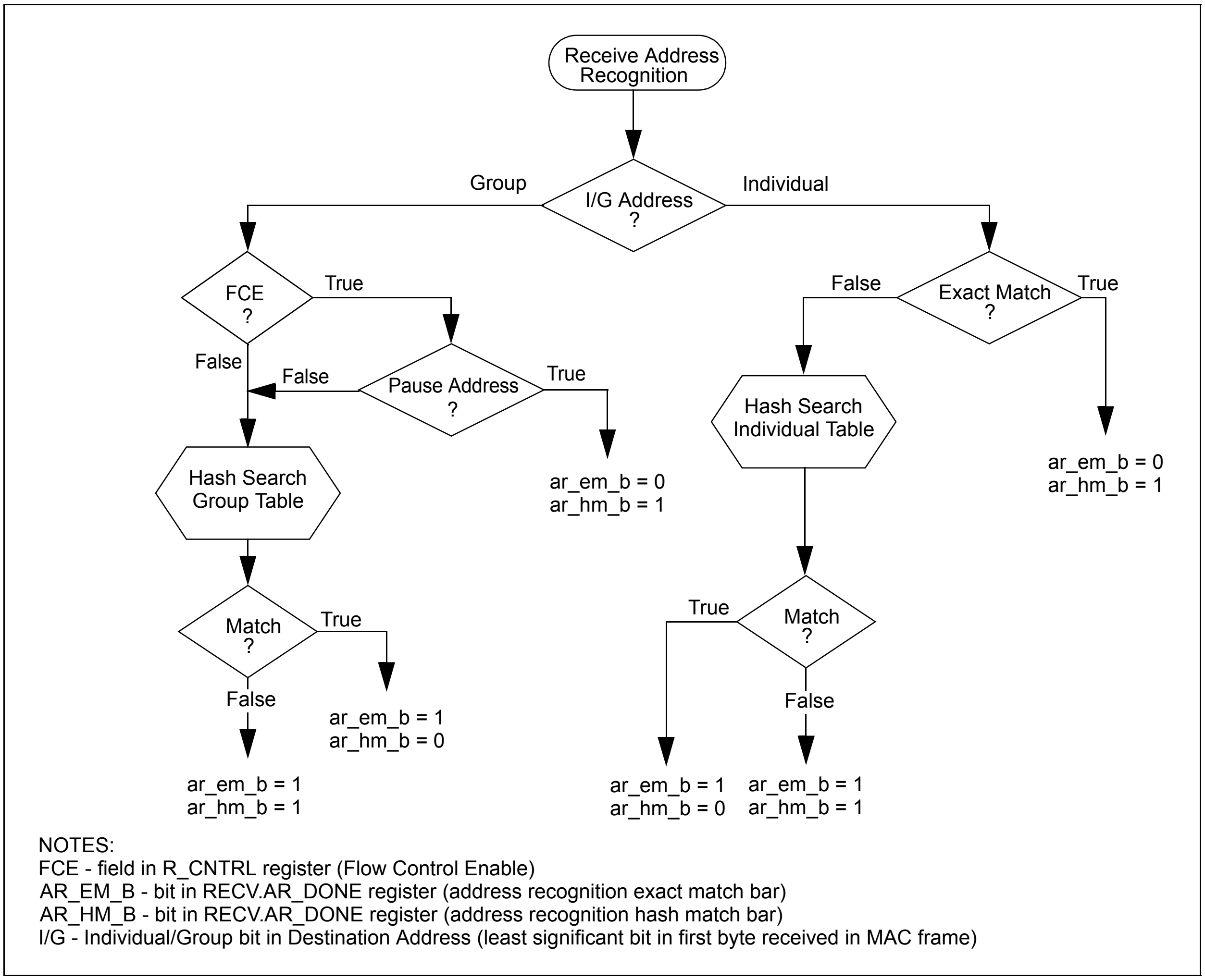


Figure 14-3. Ethernet Address Recognition - microcode decisions

The hash table algorithm used in the group and individual hash filtering operates as follows. The 48-bit destination address is mapped into one of 64 bits which are represented by 64 bits stored in GADDR1,2 (group address hash match) or IADDR1,2 (individual address hash match). This mapping is performed by passing the 48-bit address through the on-chip 32-bit CRC generator and selecting the 6 most significant bits of the CRC-encoded result to generate a number between 0 and 63. The MSB of the CRC result selects GADDR1 (MSB = 1) or GADDR2 (MSB = 0). The least significant 5 bits of the hash result select the bit within the selected register. If the CRC generator selects a bit that is set in the hash table, the frame is accepted; otherwise, it is rejected.

For example, if eight group addresses are stored in the hash table and random group addresses are received, the hash table prevents roughly 56/64 (or 87.5%) of the group address frames from reaching memory. Those that do reach memory must be further filtered by the processor to determine if they truely contain one of the eight desired addresses.

The effectiveness of the hash table declines as the number of addresses increases.

The hash table registers must be initialized by the user. The user may compute the hash for a particular address in software. The CRC32 polynomial to use in computing the hash is:

$$X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^{8} + X^{7} + X^{5} + X^{4} + X^{2} + X + 1$$

A table of example Destination Addresses and corresponding hash values is included below for reference.

Table 14-44. Destination Address to 6-Bit Hash

48-bit DA	6-bit hash (in hex)	hash decimal value
65:ff:ff:ff:ff:ff	0x0	0
55:ff:ff:ff:ff:ff	0x1	1
15:ff:ff:ff:ff:ff	0x2	2
35:ff:ff:ff:ff:ff	0x3	3
b5:ff:ff:ff:ff:ff	0x4	4
95:ff:ff:ff:ff:ff	0x5	5
d5:ff:ff:ff:ff:ff	0x6	6
f5:ff:ff:ff:ff:ff	0x7	7
db:ff:ff:ff:ff:ff	0x8	8
fb:ff:ff:ff:ff:ff	0x9	9
bb:ff:ff:ff:ff:ff	0xa	10
8b:ff:ff:ff:ff:ff	0xb	11
0b:ff:ff:ff:ff:ff	0xc	12
3b:ff:ff:ff:ff:ff	0xd	13
7b:ff:ff:ff:ff:ff	0xe	14
5b:ff:ff:ff:ff:ff	0xf	15
27:ff:ff:ff:ff:ff	0x10	16
07:ff:ff:ff:ff:ff	0x11	17
57:ff:ff:ff:ff:ff	0x12	18
77:ff:ff:ff:ff:ff	0x13	19
f7:ff:ff:ff:ff:ff	0x14	20
c7:ff:ff:ff:ff:ff	0x15	21
97:ff:ff:ff:ff:ff	0x16	22
a7:ff:ff:ff:ff:ff	0x17	23
99:ff:ff:ff:ff:ff	0x18	24
b9:ff:ff:ff:ff:ff	0x19	25
f9:ff:ff:ff:ff:ff	0x1a	26
c9:ff:ff:ff:ff:ff	0x1b	27
59:ff:ff:ff:ff:ff	0x1c	28
79:ff:ff:ff:ff:ff	0x1d	29
29:ff:ff:ff:ff:ff	0x1e	30
19:ff:ff:ff:ff:ff	0x1f	31
d1:ff:ff:ff:ff:ff	0x20	32
f1:ff:ff:ff:ff:ff	0x21	33
b1:ff:ff:ff:ff:ff	0x22	34

Table 14-44. Destination Address to 6-Bit Hash (continued)

48-bit DA	6-bit hash (in hex)	hash decimal value
91:ff:ff:ff:ff:ff	0x23	35
11:ff:ff:ff:ff:ff	0x24	36
31:ff:ff:ff:ff:ff	0x25	37
71:ff:ff:ff:ff:ff	0x26	38
51:ff:ff:ff:ff:ff	0x27	39
7f:ff:ff:ff:ff:ff	0x28	40
4f:ff:ff:ff:ff:ff	0x29	41
1f:ff:ff:ff:ff:ff	0x2a	42
3f:ff:ff:ff:ff:ff	0x2b	43
bf:ff:ff:ff:ff:ff	0x2c	44
9f:ff:ff:ff:ff:ff	0x2d	45
df:ff:ff:ff:ff:ff	0x2e	46
ef:ff:ff:ff:ff:ff	0x2f	47
93:ff:ff:ff:ff:ff	0x30	48
b3:ff:ff:ff:ff:ff	0x31	49
f3:ff:ff:ff:ff:ff	0x32	50
d3:ff:ff:ff:ff:ff	0x33	51
53:ff:ff:ff:ff:ff	0x34	52
73:ff:ff:ff:ff:ff	0x35	53
23:ff:ff:ff:ff:ff	0x36	54
13:ff:ff:ff:ff:ff	0x37	55
3d:ff:ff:ff:ff:ff	0x38	56
0d:ff:ff:ff:ff:ff	0x39	57
5d:ff:ff:ff:ff:ff	0x3a	58
7d:ff:ff:ff:ff:ff	0x3b	59
fd:ff:ff:ff:ff:ff	0x3c	60
dd:ff:ff:ff:ff:ff	0x3d	61
9d:ff:ff:ff:ff:ff	0x3e	62
bd:ff:ff:ff:ff:ff	0x3f	63

14.9.7 Full-Duplex Flow Control

Full-duplex flow control allows the user to transmit pause frames and to detect received pause frames. Upon detection of a pause frame, MAC data frame transmission stops for a given pause duration.

To enable pause frame detection, the FEC must operate in full-duplex mode (X_CNTRL.FDEN asserted) and flow control enable (R_CNTRL.FCE) must be asserted. The FEC detects a pause frame when the fields of the incoming frame match the pause frame specifications as shown in the table below. In addition, the receive status associated with the frame should indicate that the frame is valid

Table 14-45. PAUSE Frame Field Specification

48-bit destination address	0180_c200_0001 or Physical ADDRESS
48-bit Source Address	any
16-bit type	8808
16-bit opcode	0001
16-bit PAUSE duration	0000 to ffff

Pause frame detection is performed by the receiver and microcontroller modules. The microcontroller runs an address recognition subroutine to detect the specified pause frame destination address, while the receiver detects the type and opcode pause frame fields. On detection of a pause frame, graceful transmit stop is asserted by the FEC internally. When transmission has paused, the GRA (Graceful Stop complete) interrupt is asserted and the pause timer begins to increment. Note that the pause timer makes use of the transmit backoff timer hardware which is used for tracking the appropriate collision backoff time in half-duplex mode. The pause timer increments once every slot time until PAUSE_DURATION slot times have expired. On PAUSE_DURATION expiration, graceful transmit stop is deasserted allowing MAC data frame transmission to resume. Note that the receive flow control pause (X_CNTRL.RFC_PAUSE) status bit is asserted while the transmitter is paused due to reception of a pause frame.

To transmit a pause frame the FEC must operate in full-duplex mode and the user must assert flow control pause (X_CNTRL.TFC_PAUSE). On assertion of transmit flow control pause (X_CNTRL.TFC_PAUSE) the transmitter asserts graceful transmit stop internally. When the transmission of data frames stops the GRA (Graceful Stop complete) interrupt asserts. Following GRA assertion the Pause frame is transmitted. On completion of pause frame transmission flow control pause (X_CNTRL.TFC_PAUSE) and graceful transmit stop are deasserted internally.

During pause frame transmission the transmit hardware places data into the transmit data stream from the registers shown in the table below.

Table 14-46. Transmit Pause Frame Registers

PAUSE FRame fields	FEC register	Register Contents
48-bit destination address	{FDXFC_DA1[0:31], FDXFC_DA2[0:15]}	0180_c200_0001
48-bit Source Address	{PADDR1[0:31], PADDR2[0:15]}	physical address
16-bit type	PADDR2[16:31]	8808
16-bit opcode	OP_PAUSE[0:15]	0001
16-bit PAUSE duration	OP_PAUSE[16:31]	0000 to ffff

The user must specify the desired pause duration in the OP_PAUSE register.

Note that when the transmitter is paused due to receiver/microcontroller pause frame detection, transmit flow control pause (X_CNTRL.TFC_PAUSE) still may be asserted and will cause the transmission of a single pause frame. In this case the GRA interrupt will not be asserted.

14.9.8 Inter-Packet Gap Time

The minimum inter packet gap time for back-to-back transmission is 96 bit times. After completing a transmission or after the backoff algorithm completes the transmitter waits for carrier sense to be negated before starting its 96 bit time IPG counter. Frame transmission may begin 96 bit times after carrier sense is negated if it stays negated for at least 60 bit times. If carrier sense asserts during the last 36 bit times it will be ignored and a collision will occur.

The receiver receives back-to-back frames with a minimum spacing of at least 28 bit times. If an inter-packet gap between receive frames is less than 28 bit times the following frame may be discarded by the receiver.

14.9.9 Collision Handling

If a collision occurs during frame transmission the ethernet controller will continue the transmission for at least 32 bit times, transmitting a JAM pattern consisting of 32 1’s. If the collision occurs during the preamble sequence the JAM pattern will be sent after the end of the preamble sequence.

If a collision occurs within 64 byte times the retry process is initiated. The transmitter waits a random number of slot times. A slot time is 512 bit times. If a collision occurs after 64 byte times no retransmission is performed and the end of frame buffer is closed with an LC error indication.

14.9.10 Internal and External Loopback

Both internal and external loopback are supported by the ethernet controller. In loopback mode both of the FIFOs are used and the FEC actually operates in a full-duplex fashion. Both internal and external loopback are configured using combinations of the LOOP and DRT bits in the R_CNTRL register and the FDEN bit in the X_CNTRL register.

For both internal and external loopback set FDEN = 1.

For internal loopback set LOOP = 1 and DRT = 0. TX_EN and TX_ER will not assert during internal loopback. During internal loopback the transmit/receive data rate is higher than in normal operation because the internal system clock is used by the transmit and receive blocks instead of the clocks from the external transceiver. This will cause an increase in the required system bus bandwidth for transmit and receive data being transferred to/from external memory. It may be necessary to pace the frames on the transmit side and/or limit the size of the frames to prevent transmit FIFO underrun and receive FIFO overflow.

For external loopback set LOOP = 0, DRT = 0 and configure the external transceiver for loopback.

14.9.11 Ethernet Error-Handling Procedure

The ethernet controller reports frame reception and transmission error conditions using the FEC BDs (receive), the IEVENT register and the MIB block counters.

14.9.11.1 Transmission Errors

Transmitter Underrun

- If this error occurs the FEC sends 32 bits that ensure a CRC error and stops transmitting. All remaining buffers for that frame are then flushed and closed. The UN bit is set in the X_STATUS register. The FEC will then continue to the next transmit buffer descriptor and begin transmitting the next frame.
- The XFIFO_UN interrupt will be asserted if enabled in the IMASK register.

Carrier Sense Lost During Frame Transmission

- When this error occurs and no collision is detected in the frame the FEC sets the CSL bit in X_STATUS register. The frame is transmitted normally. No retries are performed as a result of this error.
- No interrupt is generated as a result of this error.

Retransmission Attempts Limit Expired

- When this error occurs the FEC terminates transmission. All remaining buffers for that frame are then flushed and closed and the RL bit is set in the X_STATUS register. The FEC will then continue to the next transmit buffer descriptor and begin transmitting the next frame.
- The COL_RETRY_LIM interrupt will be asserted if enabled in the IMASK register.

Late Collision

- When a collision occurs after the slot time (512 bits starting at the Preamble), the FEC terminates transmission. All remaining buffers for that frame are then flushed and closed and the LC bit is set in the X_STATUS register. The FEC will then continue to the next transmit buffer descriptor and begin transmitting the next frame.
- The LATE_COL interrupt will be asserted if enabled in the IMASK register.

Heartbeat

- Some transceivers have a self-test feature called “heartbeat” or “signal quality error.” To signify a good self-test the transceiver indicates a collision to the FEC within 20 clocks after completion of a frame transmitted by the Ethernet controller. This indication of a collision does not imply a real collision error on the network but is rather an indication that the transceiver still seems to be functioning properly. This is called the heartbeat condition.
- If the HBC bit is set in the X_CNTRL register and the heartbeat condition is not detected by the **FEC** after a frame transmission a heartbeat error occurs. When this error occurs the FEC closes the buffer, sets the HB bit in the X_STATUS register and generates the HBERR interrupt if it is enabled.

14.9.11.2 Reception Errors

Overrun Error

- If the receive block has data to put into the receive FIFO and the receive FIFO is full, the FEC sets the OV bit in the receive status word. All subsequent data in the frame will be discarded and subsequent frames may also be discarded until the receive FIFO is serviced by the DMA and space is made available. At this point the receive frame/status word is written into the FIFO with the OV bit isset. This frame must be discarded by the driver.

Non-Octet Error (Dribbling Bits)

— The Ethernet controller handles up to seven dribbling bits when the receive frame terminates nonoctet aligned and it checks the CRC of the frame on the last octet boundary. If there is a CRC error, then the frame nonoctet aligned (NO) error is reported in the Receive Frame Status Word . If there is no CRC error, then no error is reported.

CRC Error

— When a CRC error occurs with no dribble bits, the FEC closes the buffer and sets the CR bit in the Receive Frame Status Word. CRC checking cannot be disabled, but the CRC error can be ignored if checking is not required.

Frame Length Violation

— When the receive frame length exceeds MAX_FL bytes the BABR interrupt will be generated and the LG bit in the end of frame Receive Frame Status Word will be set. The frame is not truncated (truncation occurs if the frame length exceeds 2047 bytes).

Truncation

— When the receive frame length exceeds 2047 bytes the frame is truncated and the TR bit is set in the receive BD.

Notes

Chapter 15
Programmable Serial Controllers (PSC)

15.1 Overview

The following sections are contained in this document:

- Section 15.2, *PSC Registers—MBAR + 0x2000, 0x2200, 0x2400, 0x2600, 0x2800, 0x2C00*
- Section 15.3, *PSC Operation Modes*

The MPC5200 has 6 independent Programmable Serial Controllers (PSCs)

- PSC1 = MBAR + 0x2000
- PSC2 = MBAR + 0x2200
- PSC3 = MBAR + 0x2400
- PSC4 = MBAR + 0x2600
- PSC5 = MBAR + 0x2800
- PSC6 = MBAR + 0x2C00

Each PSC can be clocked by an internal clock source or an external clock source. In addition, each PSC module interfaces directly to the CPU and consists of the following:

- Serial Communication Channel
- Programmable Transmit (Tx) Receive (Rx) Clock Generation
- Internal Channel Control Logic
- Interrupt Control Logic

15.1.1 PSC Functions Overview

The MPC5200 provides 6 independent PSC modules, all configuration registers and internal functions are equal for each PSC. Because of pin out limitations, not all functions are available for all PSCs on every port. Table 15-1 shows, which PSC supports which mode.

In this section the name Codec mode is used as a collective term for the “normal soft Modem” the SPI and the I2S mode and the name IrDA is a common term for the Infrared interfaces SIR, MIR and FIR. The PSC can also be used to interface externally to an AC97 device, or to a device that supports a UART protocol. Figure 15-1. shows on overview about the supported functions.

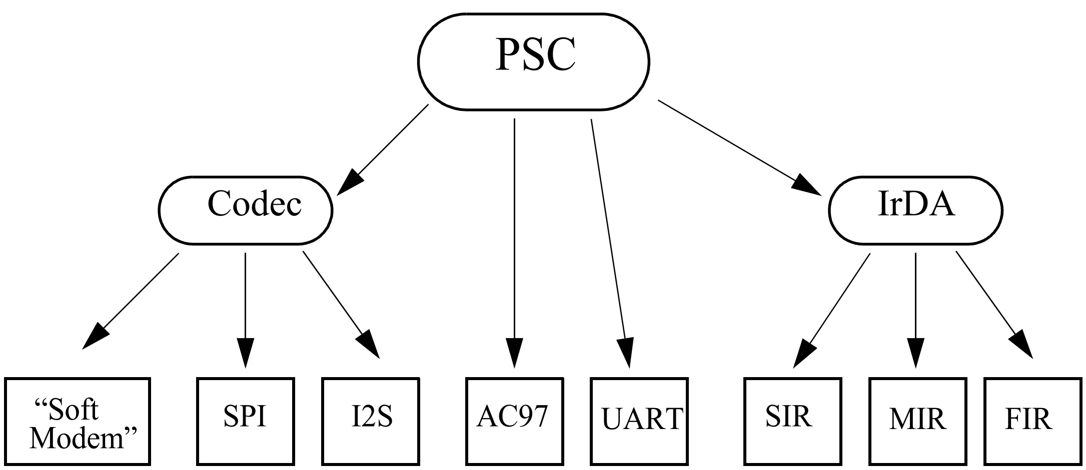


Figure 15-1. PSC Functions Overview

Table 15-1. PSC Functions Overview

	PSC1	PSC2	PSC3	PSC4	PSC5	PSC6
UART	yes	yes	yes	yes	yes	yes
Modem / SPI / I2S	yes	yes	yes	no	no	yes
Mclk Generation output	yes	yes	yes	no	no	no
AC97	yes	yes	no	no	no	no
IrDA	no	no	no	no	no	yes
Cell Phone	master	slave	slave	no	no	slave

Figure 15-2 shows a simplified PSC block diagram.

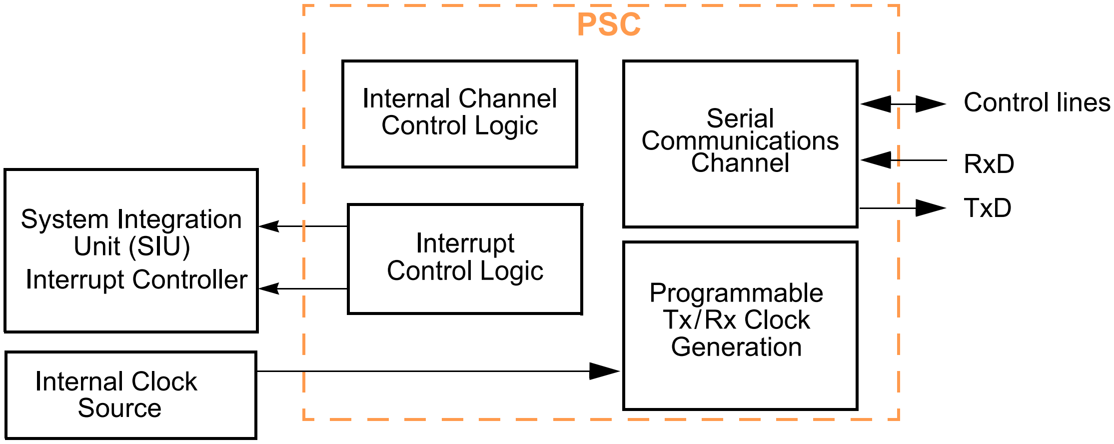


Figure 15-2. Simplified Block Diagram

The PSC can interface to a Codec through a serial port consisting of Tx and Rx serial data and serial bit-clock and frame signals. Digital sample data is transferred to and from the Codec through the serial port. The PSCs support Codec mode with 8, 16, 24 and 32 bit data width and the SPI and I2S operation modes. A Codec chip provides a data conversion interface for high-speed Modem designs meeting a high range of standards, such as ITU-T V.34 and PCM. In addition the PSC provide an Mclk for the external Codec, eliminating the need for an external crystal for the external device. For more information about the Codec mode see section: Section 15.3.2, *PSC in Codec Mode*.

AC97 defines an architecture for audio-intensive personal computer applications such as gaming, authoring, and high-resolution music and video playback. An external AC97 analog device performs mixing, analog processing, and sample-rate DAC and ADC. PSC1 and PSC2 can interface to the AC97 device through a serial port consisting of Tx and Rx serial data, a serial bit-clock input, and a frame sync output generated by the PSC from the serial bit-clock. An MPC5200 General-Purpose I/O (GPIO) is used to reset the AC97 device. The PSC transfers digital sample data as well as control/status information to and from the AC97 device through the serial port. For more information about the AC97 mode see section: Section 15.3.3, *PSC in AC97 Mode*.

When programmed as a UART the PSC serial communication channel provides a full-duplex asynchronous receiver and transmitter deriving an operating frequency from an internal clock. The transmitter converts parallel data from the CPU to a serial bit-stream, inserting appropriate start, stop, and parity bits. It outputs the resulting stream on the channel transmitter serial data output (TxD). The receiver converts serial data from the channel receiver serial data input (RxD) to parallel format, checks for start, stop, and parity bits, or line break conditions, and transfers the assembled character onto the bus during read operations. The receiver may be poll-driven or interrupt-driven. For more information about the UART mode see section: Section 15.3.1, *PSC in UART Mode*.

15.1.2 Features

General Features:

- 512-byte receiver (Rx) FIFO
- 512-byte transmitter (Tx) FIFO
- Each channel is programmable to normal (full-duplex), automatic echo, local loop-back, or remote loop-back mode
- Automatic Walk-up mode for multidrop applications
- 6 maskable interrupt conditions
- PSC Tx and Rx FIFOs can be programmed to interrupt either the BestComm or the CPU when they require filling or emptying, respectively.

PSC UART mode:

- Each is clocked by an internal clock source (IPB clock), eliminating the need for an external crystal
- Full-duplex asynchronous/synchronous receiver/transmitter channel
- Programmable data format:
  - — five to eight data bits plus parity
  - — Odd, even, no parity, or force parity
  - — One, one-and-a-half, or two STOP bits
- Parity, framing, and overrun error detection

- False-start bit detection
- Line-break detection and generation
- Detection of breaks originating in the middle of a character
- Start/end break interrupt/status

PSC Codec mode:

- Programmable to interface to an 8, 16, 24 or 32bit Codec for “soft modem” support
- Support master mode, driving clock and frame sync signals
- Support slave mode, receiving clock and the frame sync from the external Codec
- Supports full duplex SPI interface
- Supports I2S interface
- No parity error, framing error, or line break detection in Codec mode
- Ability to generate a master clock (Mclk) for an external Codec device, independent from the mode (master or slave)
- Programmable width of the frame sync signal
- Frame sync and bit clock frequencies are independently programmable
- Frame sync and bit clock polarity are programmable
- Support “digital cell phone” interface

AC97 mode:

- PSC1 and PSC2 support an AC97 interface

IrDA SIR mode:

- Baud rate: 2400 to 115200 bps
- Selectable pulse width: either 3/16 bit duration or 1.6 µs

IrDA MIR mode:

- Baud rate: 0.576 Mbps to 1.152 Mbps

IrDA FIR mode:

- Baud rate: 4 Mbps

15.2 PSC Registers—MBAR + 0x2000, 0x2200, 0x2400, 0x2600, 0x2800, 0x2C00

The PSCs are located at an address as indicated below:

- PSC1 BASE = MBAR + 0x2000
- PSC2 BASE = MBAR + 0x2200
- PSC3 BASE = MBAR + 0x2400
- PSC4 BASE = MBAR + 0x2600
- PSC5 BASE = MBAR + 0x2800
- PSC6 BASE = MBAR + 0x2C00

Each PSC uses 42 registers. The register address is calculated as base address for the regarding PSC plus the offset value. Table 15-2 shows the list with all implemented registers and the associated offset value.

Table 15-2. PSC Memory Map

Offset	Register Name	Register width	Access
00	Mode Register 1 (0x00)—MR1	8	R/W
00	Mode Register 2 (0x00) — MR2	8	R/W
04	Status Register (0x04) — SR	16	R
04	Clock Select Register (0x04) — CSR	16	W
08	Command Register (0x08)—CR	8	R/W
0C	Rx Buffer Register (0x0C) — RB	32	R
0C	Tx Buffer Register (0x0C)—TB	32	W

Table 15-2. PSC Memory Map (continued)

10	Input Port Change Register (0x10) — IPCR	8	R
10	Auxiliary Control Register (0x10) — ACR	8	W
14	Interrupt Status Register (0x14) — ISR	16	R
14	Interrupt Mask Register (0x14)—IMR	16	W
18	Counter Timer Upper Register (0x18)—CTUR	8	W
1C	Counter Timer Lower Register (0x1C)—CTLR	8	W
20	Codec Clock Register (0x20)—CCR	16	R/W
30	Interrupt Vector Register (0x30)—IVR - Reserved	8	R/W
34	Input Port Register (0x34)—IP	8	R
38	Output Port 1 Bit Set (0x38)—OP1	8	W
3C	Output Port 0 Bit Set (0x3C)—OP0	8	W
40	Serial Interface Control Register (0x40)—SICR	32	R/W
44	Infrared Control 1 (0x44)—IRCR1	8	R/W
48	Infrared Control 2 (0x48)—IRCR2	8	R/W
4C	Infrared SIR Divide Register (0x4C)—IRSDR	8	R/W
50	Infrared MIR Divide Register (0x50)—IRMDR	8	R/W
54	Infrared FIR Divide Register (0x54)—IRFDR	8	R/W
58	Rx FIFO Number of Data (0x58)—RFNUM	16	R
5C	Tx FIFO Number of Data (0x5C)—TFNUM	16	R
60	Rx FIFO Data (0x60)—RFDATA	32	R/W
64	Rx FIFO Status (0x64)—RFSTAT	16	R/W
68	Rx FIFO Control (0x68)—RFCNTL	8	R/W
6E	Rx FIFO Alarm (0x6E)—RFALARM	16	R/W
72	Rx FIFO Read Pointer (0x72)—RFRPTR	16	R/W
76	Rx FIFO Write Pointer(0x76)—RFWPTR	16	R/W
7A	Rx FIFO Last Read Frame (0x7A)—RFLRFPTR - Reserved	16	R/W
7C	Rx FIFO Last Write Frame PTR (0x7C)—RFLWFPTR - Reserved	16	R/W
80	Tx FIFO Data (0x80)—TFDATA	32	R/W
84	Tx FIFO Status (0x84)—TFSTAT	16	R/W
88	Tx FIFO Control (0x88)—TFCNTL	8	R/W
8E	Tx FIFO Alarm (0x8E)—TFALARM	16	R/W
92	Tx FIFO Read Pointer (0x92)—TFRPTR	16	R/W
96	Tx FIFO Write Pointer (0x96)—TFWPTR	16	R/W
9A	Tx FIFO Last Read Frame (0x9A)—TFLRFPTR - Reserved	16	R/W
9C	Tx FIFO Last Write Frame PTR (0x9C)—TFLWFPTR - Reserved	16	R/W

PSC module operation is controlled by writing control bytes into the appropriate registers.

15.2.1 Mode Register 1 (0x00)—MR1

The Mode registers control configuration. MR1 can be read or written when the Mode register pointer points to it, at reset or after a reset Mode register pointer command using CR[MISC]. After MR1 is read or written, the pointer points to MR2.

Table 15-3. Mode Register 1 (0x00) for UART Mode

	msb 0	1	2	3	4	5	6	7 lsb
R / W	RxRTS	RxIRQ/FFULL	Reserved	PM		PT	B/C	
RESET:	0	0	1	0	0	0	0	0

Table 15-4. Mode Register 1 (0x00) for SIR Mode

	msb 0	1	2	3	4	5	6	7 lsb
R / W	RxRTS	RxIRQ/FFULL	Reserved					
RESET:	0	0	1	1	0	0	1	1

Table 15-5. Mode Register 1 (0x00) for other Modes

	msb 0	1	2	3	4	5	6	7 lsb
R / W	Reserved	RxIRQ/FFULL	Reserved					
RESET:	0	0	1	1	0	0	1	1

Bit	Name	Description
0	RxRTS	**UART / SIR**—Receiver request-to-send—Allows $\overline{RTS}$ output to control the $\overline{CTS}$ input of the transmitting device to prevent receiver overrun. If both the receiver and transmitter are incorrectly programmed for $\overline{RTS}$ control, $\overline{RTS}$ control is disabled for both. Transmitter RTS control is configured in MR2[TxRTS]. Not used in Codec mode. 0 = Receiver has no effect on $\overline{RTS}$. 1 = When a valid start bit is received, $\overline{RTS}$ is negated if the PSC FIFO is full. $\overline{RTS}$ is reasserted when the FIFO has an empty position available. **other Modes**—Reserved
1	RxIRQ/ FFULL	Receiver interrupt select. 0 = RxRDY is the source that generates IRQ 1 = FFULL is the source that generates IRQ.
2	—	Reserved
3:4	PM	**UART**—Parity mode—Selects the parity or multidrop mode for the channel. The parity bit is added to the transmitted character, and the receiver performs a parity check on incoming data. The value of PM affects PT, as shown Table 15-6. PM is not used in Codec mode. **other Modes**—Reserved

Bit	Name	Description
5	PT	**UART**—Parity Type—PM and PT together select parity type (PM = 0x) or determine whether a data or address character is transmitted (PM = 11). PT is not used in Codec mode. See Table 15-6. **other Modes**—Reserved
6:7	B/C	**UART**—Bits per Character—Select the number of data bits per character to be sent. The values shown do not include start, parity, or stop bits. B/C is not used in Codec mode. 00 = 5 bits 01 = 6 bits 10 = 7 bits 11 = 8 bits **other Modes**—Reserved

Table 15-6. Parity Mode/Parity Type Definitions

PM	Parity Mode	Parity Type (PT=0)	Parity Type (PT=1)
00	With parity	Even parity	Odd parity
01	Force parity	Low parity	High parity
10	No parity	n/a	
11	Multidrop mode	Data character	Address character

15.2.2 Mode Register 2 (0x00) — MR2

MR2 can be read or written when the Mode register pointer points to it, which occurs after any access to MR1. An MR2 access does not update the mode register address.

Table 15-7. Mode Register 2 (0x00) for UART / SIR Mode

	msb 0	1	2	3	4	5	6	7 lsb
R	CM		TxRTS	TxCTS	SB			
W								
RESET:	0	0	0	0	0	0	0	0

Table 15-8. Mode Register 2 (0x00) for other Modes

	msb 0	1	2	3	4	5	6	7 lsb
R	CM		Reserved					
W								
RESET:	0	0	0	0	0	0	0	0

Bit	Name	Description
0:1	CM	Channel mode—Selects a channel mode.CM is used in both UART and Codec modes. 00 = Normal 01 = Automatic echo 10 = Local loop-back 11 = Remote loop-back
2	TxRTS	**UART / SIR**—Transmitter ready-to-send—Controls negation of $\overline{RTS}$ to automatically terminate a message transmission. Attempting to program a receiver and transmitter in the same channel for $\overline{RTS}$ control is not permitted and disables $\overline{RTS}$ control for both. TxRTS is not used in Codec mode. 0 = The transmitter has no effect on $\overline{RTS}$. 1 = In applications where the transmitter is disabled after transmission completes, setting this bit automatically clears RTS line one bit-time after any characters in the channel transmitter shift and holding registers are completely sent, including the programmed number of stop bits. **other Modes**—Reserved
3	TxCTS	**UART / SIR**—Transmitter clear-to-send—If both TxCTS and TxRTS are enabled, TxCTS controls the operation of the transmitter. TxCTS is not used in Codec mode. 0 = $\overline{CTS}$ has no effect on the transmitter. 1 = Enables clear-to-send operation. The transmitter checks the state of $\overline{CTS}$ each time it is ready to send a character. If $\overline{CTS}$ is asserted, the character is sent If it is negated, the channel TxD remains in a high state and transmission is delayed until $\overline{CTS}$ is asserted. Changes in $\overline{CTS}$ as a character is being sent do not affect its transmission. **other Modes**—Reserved
4:7	SB	**UART**—Stop-Bit (length control)—Selects the stop bit length that is appended to the transmitted character. Stop-bit lengths of 9/16th to 2 bits are programmable for 6-, 8-bit characters. Lengths of 1 1/16th to 2 bits are programmable for 5-bit characters. In all cases, the receiver checks only for a high condition at the center of the first stop-bit position, that is, one bit-time after the last data bit or after the parity bit, if parity is enabled. If an external 1x clock is used for the transmitter, clearing bit 3 selects 1 stop bit and setting bit 3 selects 2 stop bits for transmission. Not used in Codec mode, see Table 15-9. **other Modes**—Reserved

Table 15-9. Stop-Bit Lengths

SB	5 Bits	6–8 Bits	SB	5 Bits	6–8 Bits	SB	5–8 Bits	SB	5–8 Bits
0000	1.063	0.563	0100	1.313	0.813	1000	1.563	1100	1.813
0001	1.125	0.625	0101	1.375	0.875	1001	1.625	1101	1.875
0010	1.188	0.688	0110	1.438	0.938	1010	1.688	1110	1.938
0011	1.250	0.750	0111	1.500	1.000	1011	1.750	1111	2.000

15.2.3 Status Register (0x04) — SR

The read-only SR register shows status of the transmitter, the receiver, and the FIFO.

Table 15-10. Status Register (0x04) for UART Mode

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15 lsb
R	RB	FE	PE	ORERR	TxEMP	TxRDY	FFULL	RxRDY	CDE	Reserved						
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 15-11. Status Register (0x04) for SIR Mode

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15 lsb
R	RB	FE	PE	ORERR	TxEMP	TxRDY	FFULL	RxRDY	Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 15-12. Status Register (0x04) for MIR / FIR Mode

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15 lsb
R	EOF	FHYERR	PE	ORERR	URERR	TxRDY	FFULL	RxRDY	DEOF	Reserved						
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 15-13. Status Register (0x04) for other Modes

	msb 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15 lsb
R	Reserved			ORERR	URERR	TxRDY	FFULL	RxRDY	Reserved							
W																
RESET:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Name	Description
0	RB/ EOF	**UART / SIR** —Received Break—detects breaks originating in middle of received character. Such a break must persist until the end of next detected character time. 0 = No break received. 1 = An all-0 character of the programmed length was received without a stop bit. RB is valid only when RxRDY = 1. Only a single FIFO position is occupied when a break is received. Further entries to FIFO are inhibited until RxD returns to high state for at least one-half bit-time, which equals two successive PSC clock edges. MIR / SIR—End of frame 0 = The next byte to be read from the RX-FIFO is not the last one of the frame. 1= The next byte to be read from the RX-FIFO is the last one of the frame. This bit is effective when RxRDY=1. **other Modes**—Reserved
1	FE/ PHYERR	**UART / SIR**—Framing Error— is not used (always 0) in Codec mode. 0 = No framing error occurred. 1 = No stop bit detected when corresponding FIFO data character received. Stop bit-check occurs in middle of first stop bit position. FE is valid only when RxRDY=1. MIR / FIR—Physical layer error 0 = No error 1 = In MIR mode, this denotes that the RX received an abort. In FIR mode, this denotes that there was a decode error. This bit can be cleared by the reset error status command in the CR. **other Modes**—Reserved
2	PE	**UART / SIR**—Parity Error—valid only if RxRDY = 1. PE is not used (always 0) in Codec mode. 0 = No parity error occurred. 1 = If MR1[PM]=0x (with parity or force parity), corresponding FIFO character was received with incorrect parity. If MR1[PM]=11 (multidrop), PE stores received A/D bit.\ **other Modes**—Reserved
3	ORERR	Overrun Error Indicates whether an overrun occurs. For purposes of overrun, FIFO full means all FIFO space is occupied; the Rx FIFO threshold is irrelevant to overrun. 0 = No overrun occurred. 1 = One or more characters in Rx data stream were lost. ORERR sets on receipt of a new character when FIFO is full and a character is already in the shift register waiting for an empty FIFO position. When this occurs, the character in the Rx shift register and its break detect, framing error status, and parity error, if any, are lost. ORERR is cleared by the RESET ERROR STATUS command in CR. Also see the note on the end of this table.

Bit	Name	Description
4	TxEMP/URERR	**UART / SIR**—Transmitter Empty 0 = Tx buffer not completely empty. Either a character is being shifted out, or Tx is disabled. Tx is enabled/disabled by programming CR [TC]. 1 = Tx has underrun (both the Tx holding register and Tx shift registers are empty). This bit sets after transmission of the last stop bit of a character, if there are no characters in the Tx holding register awaiting transmission. **other Modes**—Underrun error 0 = No error. 1 = Underrun error occurred, which means the number of Tx FIFO bytes is 0, the Tx shift register is empty, and a frame sync occurs. In other words, the time has come to transmit a new sample, but no sample is available in the Tx shift register. Unlike UART mode, TxEMP high indicates an error condition similar to the overrun condition (ORERR = 1), and as such it is now cleared the same way as ORERR, by a RESET ERROR STATUS command in the CR and not by a reset Tx command in the CR. Also see the note on the end of this table.
5	TxRDY	Transmitter Ready 0 - Tx FIFO contains a number of data bytes greater than the TFALARM register value, or the Tx is disabled. 1 - Tx FIFO is “almost empty” as defined by the TFALARM. TxRDY sets when the number of Tx FIFO bytes falls to, or below, the TFALARM value, due to data transfer from the Tx FIFO to the Tx shift register. Once set, TxRDY remains set until the number of empty bytes in the Tx FIFO falls to 4 times the granularity level specified in the TFCNTL register. In UART mode this bit only asserts if the Tx is enabled. Also see the note on the end of this table.
6	FFULL	Rx FIFO full 0 = The Rx FIFO is not “almost full” 1 = Rx FIFO is “almost full” as defined by the RFALARM. FFULL sets as soon as the number of bytes in the Rx FIFO exceeds the RFALARM value, due to the transfer of data from the Rx shift register to the Rx FIFO. Once set, FFULL remains set until the number of bytes in the Rx FIFO falls to the granularity level specified in the RFCNTL register. Also see the note on the end of this table.
7	RxRDY	Receiver Ready 0 = There is no data in the Rx FIFO. 1 = One or more characters were received and are waiting in the Rx buffer FIFO. Also see the note on the end of this table.
8	CDE	**UART**—DCD Error 0 = The $\overline{\text{DCD}}$ input is negated while receiving data. 1 = No error **other Modes**—Reserved
9:15	—	Reserved

NOTE

The FIFO related status bits ORERR, URERR, RxRDY, FFUL and TxRDY will be changed only if the peripheral (transmitter or receiver) access the FIFO. These bits reflect to the related bits in the ISR, therefore only the peripheral side can generate a FIFO access interrupt. The bits in the RFSTAT or TFSTAT register are also set. If the CPU side read from an empty FIFO or write to a full FIFO the status bits in the RFSTAT or TFSTAT register will be set, but the status bit in the SR register are unchanged. An access from the CPU side to the FIFO can’t generate an interrupt.